MOBILE AD-HOC NETWORKS

Ad-Hoc Basic Concepts – Characteristics – Applications – Design Issues – Routing – Essential of Traditional Routing Protocols – Popular Routing Protocols – Vehicular Ad-Hoc networks (VANET) – MANET Vs VANET – Security

*Prasant Kumar Pattnaik, Rajib Mall, “Fundamentals of Mobile Computing”, PHI Learning Pvt. Ltd, New Delhi

Synopsis

- Ad-Hoc Basic Concepts
- Characteristics of MANETs
- Applications of MANETs
- MANET Design Issues
- Routing
- Essential of Traditional Routing Protocols
- Popular Routing Protocols
- Vehicular Ad-Hoc networks (VANET)
- MANET Vs VANET
- Security

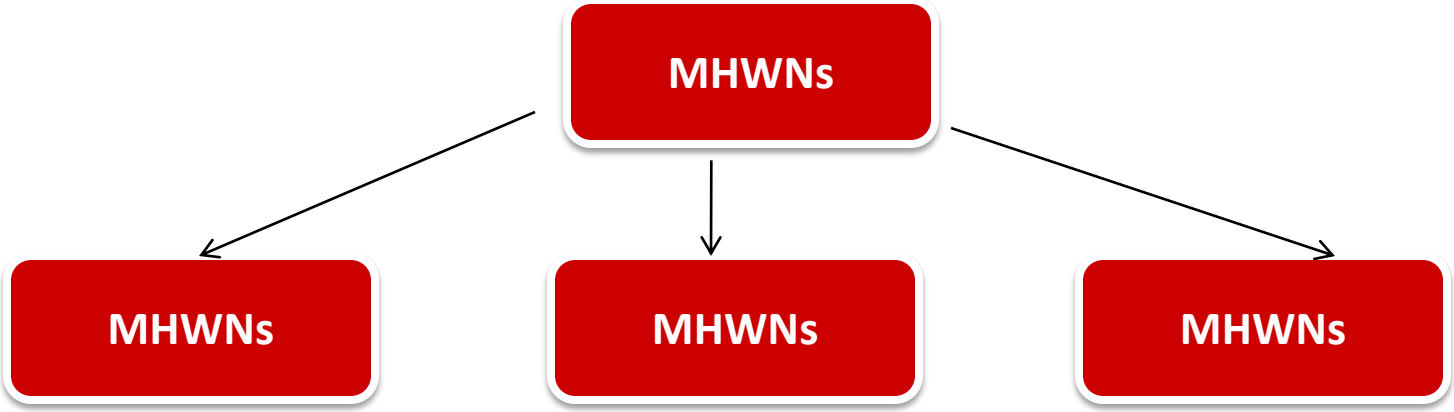
Ad-Hoc Basic Concepts

Multi-hop Wireless Networks (MHWNs)

- It is defined as a collection of nodes that communicate with each other wirelessly by using radio signals with a shared common channel.
- which forms a temporary network without the aid of centralized administration or standard support devices regularly available as conventional networks.
- There are several names for MHWNs; it could be called packet radio network, Ad-Hoc network or mobile network.

Ad-Hoc Basic Concepts

Multi-hop Wireless Networks (MHWNs)



- A mobile ad hoc network (MANET) is a continuously self-configuring, infrastructure-less network of mobile devices connected without wires.
- A mobile ad-hoc network (MANET) is an ad-hoc network but an ad-hoc network is not necessarily a MANET.

Types of MANET

- Vehicular Ad hoc Networks (VANETs)
- Smart Phone Ad hoc Networks (SPANs)
- Internet based mobile ad hoc networks (iMANETs)
- Military / Tactical MANETs

Characteristics of MANET

- Autonomous and infrastructureless
- Multi-hop routing
- Dynamic network topology
- Device heterogeneity
- Energy constrained operation
- Bandwidth constrained variable capacity links
- Limited physical security
- Network scalability
- Self-creation, self-organization and self-administration

Applications of MANET

- Tactical networks
- Emergency services
- Commercial and civilian environments
- Home and enterprise networking
- Education
- Entertainment
- Sensor networks
- Context aware services
- Coverage extension

MANET Design Issues

- Unpredictability of Environment
- Unreliability of Wireless Medium
- Resource-Constrained Nodes
- Dynamic Topology
- Transmission Errors
- Node Failures
- Link Failures
- Route Breakages
- Congested Nodes or Links

Routing

- To find and maintain routes between nodes in a dynamic topology with possibly uni-directional links, using minimum resources.
- Routing Protocols
  - Proactive protocols (table-based)
    - Traditional distributed shortest-path protocols.
    - Based on periodic updates. High routing overhead.

Routing

- Reactive (on-demand) protocols
  - Discover routes whenever needed to reduce routing overhead.
    - No route created a priori. It is created in response to a need. But introduces delay.
  - Source-initiated route discovery.
- Hybrid protocols

Routing Algorithms

- Objective of routing algorithms is to calculate 'good' routes
- Routing algorithms for both datagrams and virtual circuits should satisfy:
  - Correctness
  - Simplicity
  - Robustness
  - Stability
  - Optimality
  - Fairness
- Impossible to satisfy everything at the same time

Elements of Routing Algorithms

- Optimization Criteria
  - Number of Hops
  - Cost
  - Delay
  - Throughput
- Decision Time
  - Once per session (VCs)
  - Once per packet (datagram)
- Decision Place
  - Each node (distributed routing)
  - Central node (centralized routing)
  - Sending node (source routing)

Shortest-Path Routing

- Routing algorithms generally use a shortest path algorithm to calculate the route with the least cost. Three components
- Measurement Component: Nodes (routers) measure the current characteristics such as delay, throughput, and "cost"
- Protocol: Nodes disseminate the measured information to other nodes
- Calculation: Nodes run a least-cost routing algorithm to recalculate their routes

Traditional Routing Protocols

- There are two basic approaches to least-cost routing in a communication network
- There are two basic approaches to shortest-path routing
  - Link State Routing
  - Distance Vector Routing

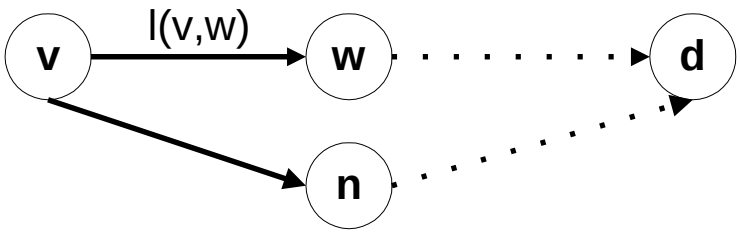
Approaches to Shortest Path Routing

- Link State Routing
  - Each node knows the distance to its neighbors
  - The distance information (=link state) is broadcast to all nodes in the network
  - Each node calculates the routing tables independently
- Distance Vector Routing
  - Each node knows the distance (=cost) to its directly connected neighbors
  - A node sends a list to its neighbors with the current distances to all nodes
  - If all nodes update their distances, the routing tables eventually converge

Distance Vector

- Each node maintains two tables:
  - Distance Table: Cost to each node via each outgoing link
  - Routing Table: Minimum cost to each node and next hop node
- Nodes exchange messages that contain information on the cost of a route
- Reception of messages triggers recalculation of routing table

Distance Vector Algorithm: Tables



Distance Table

via / to	w	n
d	$C_d(v,w)$	$C_d(v,n)$

RoutingTable

to	via (next hop)	cost
d	n	$D_d(v)$

Note: In the figure, $C_d(v,w) < C_d(v,n)$ and, therefore, $D_d(v) = C_d(v,n)$

$l(v,w)$	cost of link (w,v)
$C_d(v,w)$	cost from v to d via w
$D_d(v)$	minimum cost from v to d

Distance Vector Routing

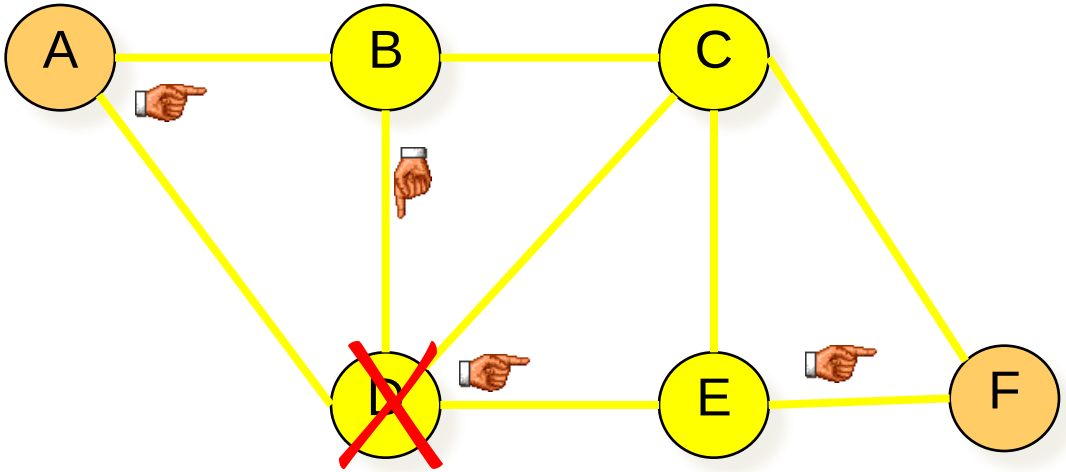
- Entries of routing tables can change while a packet is being transmitted. This can lead to a single datagram visiting the same node more than once (Looping)
- If the period for updating the routing tables is too short, routing table entries are changed before convergence (from the previous updates) is achieved
- Example: The ARPANET used a Distance Vector algorithm with an update period of <1 sec. Due to the instability of routing, the ARPANET switched in 1979 to a link state routing algorithm

Characteristics of DV Routing

- Periodic Updates: Updates to the routing tables are sent at the end of a certain time period. A typical value is 90 seconds.
- Triggered Updates: If a metric changes on a link, a router immediately sends out an update without waiting for the end of the update period.
- Full Routing Table Update: Most distance vector routing protocol send their neighbors the entire routing table (not only entries which change).
- Route invalidation timers: Routing table entries are invalid if they are not refreshed. A typical value is to invalidate an entry if no update is received after 3-6 update periods.

Distance Vector vs Link State Routing

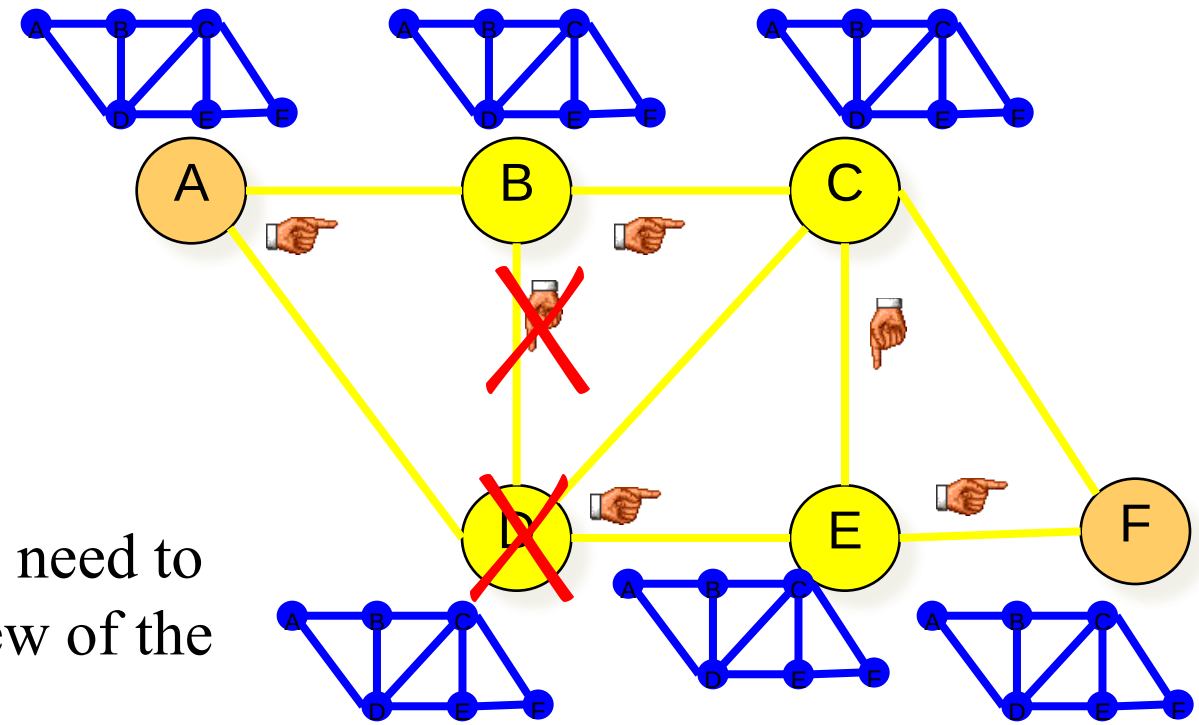
- With distance vector routing, each node has information only about the next hop
  - Node A: to reach F go to B
  - Node B: to reach F go to D
  - Node D: to reach F go to E
  - Node E: go directly to F
- Distance vector routing makes poor routing decisions if directions are not completely correct (e.g., because a node is down).



- If parts of the directions incorrect, the routing may be incorrect until the routing algorithms has re-converged.

Distance Vector vs. Link State Routing

- In link state routing, each node has a complete map of the topology
- If a node fails, each node can calculate the new route
- Difficulty: All nodes need to have a consistent view of the network



Link State Routing

- Each node must
  - discover its neighbors
  - measure the delay (=cost) to its neighbors
  - broadcast a packet with this information to all other nodes
  - compute the shortest paths to every other router
- The broadcast can be accomplished by flooding
- The shortest paths can be computer with Dijkstra's algorithm

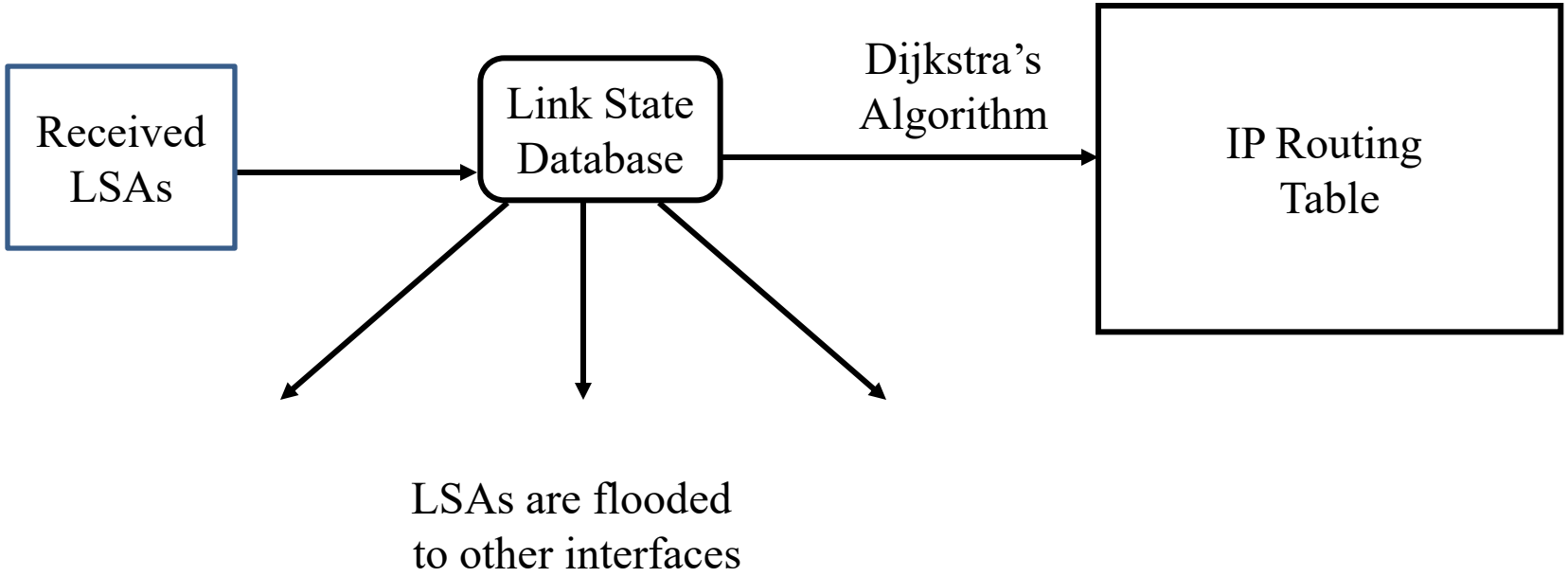
Link State Routing: Basic principles

- Each router establishes a relationship ("adjacency") with its neighbors
- Each router generates link state advertisements (LSAs) which are distributed to all routers
  - LSA = (link id, state of the link, cost, neighbors of the link)
- Each router maintains a database of all received LSAs (topological database or link state database), which describes the network has a graph with weighted edges
- Each router uses its link state database to run a shortest path algorithm (Dijikstra's algorithm) to produce the shortest path to each network

Link State Routing: Properties

- Each node requires complete topology information
- Link state information must be flooded to all nodes
- Guaranteed to converge

Operation of a LSR Protocol



Dijkstra's Shortest Path Algorithm

Input: **Graph (N,E) with**

N	**the set of nodes and $E \subseteq N \times N$ the set of edges**
d_{vw}	**link cost (d_{vw} = infinity if $(v,w) \notin E$, $d_{vv} = 0$)**
s	**source node.**

Output: **D_n** **cost of the least-cost path from node s to node n**

```
M = {s};
for each  n ∉ M
        D_n  = d_sn;
while (M ≠ all nodes) do
        Find w ∉ M for which D_w = min{D_j ; j ∉ M};
        Add w to M;
        for each  n ∉ M
                D_n = min_w [ D_n, D_w + d_wn ];
                Update route;
enddo
```

MANET vs. Traditional Routing

- Every node is potentially a router in a MANET, while most nodes in traditional wired networks do not route packets
- Nodes transmit and receive their own packets and, also, forward packets for other nodes
- Topologies are dynamic in MANETs due to mobile nodes, but are relatively static in traditional networks
- Routing in MANETs must consider both Layer 3 and Layer 2 information, while traditional protocols rely on Layer 3 information only
- Link layer information can indicate connectivity and interference

MANET vs. Traditional Routing

- MANET topologies tend to have many more redundant links than traditional networks.
- A MANET "router" typically has a single interface, while a traditional router has an interface for each network to which it connects
  - Routed packet sent forward when transmitted, but also sent to previous transmitter
- Channel properties, including capacity and error rates, are relatively static in traditional networks, but may vary in MANETs

MANET vs. Traditional Routing

- Interference is an issue in MANETs, but not in traditional networks
- Channels can be asymmetric with some Layer 2 technologies
  - Note that the IEEE 802.11 MAC assumes symmetric channels
- Power efficiency is an issue in MANETs, while it is normally not an issue in traditional networks
- MANETs may have gateways to fixed network, but are typically stub networks, while traditional networks can be stub networks or transit networks

MANET vs. Traditional Routing

- There is limited physical security in a MANET compared to a traditional network
  - Increased possibility of eavesdropping, spoofing, and denial-of-security attacks
- Traditional routing protocols for wired networks do not work well in most MANETs
  - MANETs are too dynamic
  - Wireless links present problems of interference, limited capacity, etc.

Classification of Unicast Routing

- Proactive Routing Protocol.
  - Eg. OLSR, FSR, WRP,
- Reactive Routing Protocol.
  - Eg. AODV, DSR
- Hybrid Routing Protocol
  - Eg. TORA, ZRP

Popular Routing Protocols

- Optimized Link State Routing Protocol (OLSR)
- **Destination Sequence Distance Vector(DSDV)**
- **Ad hoc On-demand Distance Vector(AODV)**
- **Dynamic Source Routing(DSR)**
- Flow-state in DSR
- Power-Aware DSR-based
- Cluster Based Routing Protocol
- Fisheye State Routing protocol
- **Zone-based Hierarchical Link State Routing Protocol**

Destination Sequence Distance Vector

- DSDV is Proactive (Table Driven)
  - Each node maintains routing information for all known destinations
  - Routing information must be updated periodically
  - Traffic overhead even if there is no change in network topology
  - Maintains routes which are never used
- Keep the simplicity of Distance Vector

Destination Sequence Distance Vector

- Guarantee Loop Freeness
  - New Table Entry for Destination Sequence Number
- Allow fast reaction to topology changes
  - Make immediate route advertisement on significant changes in routing table
  - but wait with advertising of unstable routes (damping fluctuations)

DSDV (Table Entries)

- **Sequence number** originated from destination. Ensures loop freeness.

- **Install Time** when entry was made (used to delete stale entries from table)

- **Stable Data** Pointer to a table holding information on how stable a route is. Used to damp fluctuations in network.

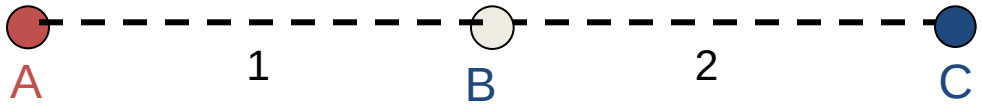
Destination	Next	Metric	Seq. Nr	Install Time	Stable Data
A	A	0	A-550	001000	Ptr_A
B	B	1	B-102	001200	Ptr_B
C	B	3	C-588	001200	Ptr_C
D	B	4	D-312	001200	Ptr_D

DSDV (Route Advertisements)

- Advertise to each neighbor own routing information
  - Destination Address
  - Metric = Number of Hops to Destination
  - Destination Sequence Number
- Rules to set sequence number information
  - On each advertisement increase own destination sequence number (use only even numbers)
  - If a node is no more reachable (timeout) increase sequence number of this node by 1 (odd sequence number) and set metric = ∞

DSDV (Route Selection)

- Update information is compared to own routing table
  - Select route with higher destination sequence number (This ensure to use always newest information from destination)
  - Select the route with better metric when sequence numbers are equal.



Dest.	Next	Metric	Seq
A	A	0	A-550
B	B	1	B-100
C	B	3	C-586

Dest.	Next	Metric	Seq
A	A	1	A-550
B	B	0	B-100
C	C	2	C-588

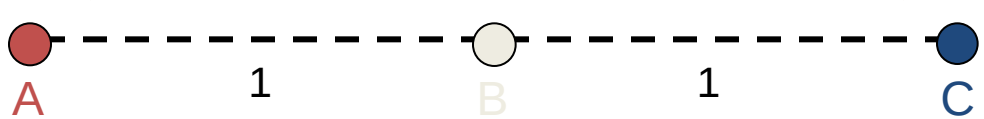
Dest.	Next	Metric	Seq.
A	B	1	A-550
B	B	2	B-100
C	C	0	C-588

DSDV (Route Advertisement)

B increases Seq.Nr from 100 -> 102
B broadcasts routing information
to Neighbors A, C including
destination sequence numbers

(A, 1, A-500)
(B, 0, B-102)
(C, 1, C-588)

(A, 1, A-500)
(B, 0, B-102)
(C, 1, C-588)



Dest.	Next	Metric	Seq
A	A	0	A-550
B	B	1	B-102
C	B	2	C-588

Dest.	Next	Metric	Seq
A	A	1	A-550
B	B	0	B-102
C	C	1	C-588

Dest.	Next	Metric	Seq.
A	B	2	A-550
B	B	1	B-102
C	C	0	C-588

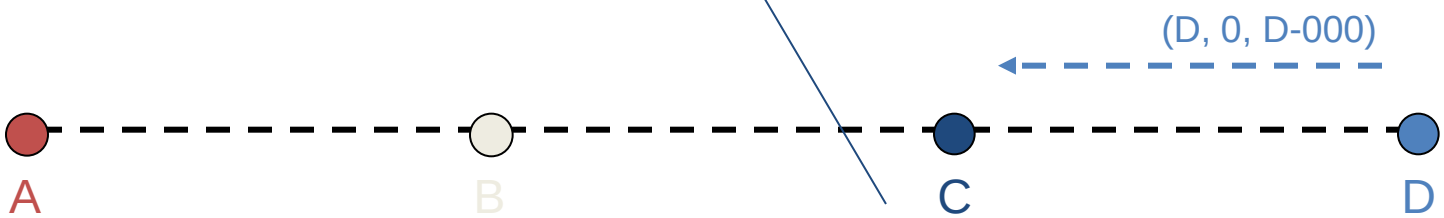
DSDV (Respond to Topology Changes)

- Immediate advertisements
  - Information on new Routes, broken Links, metric change is immediately propagated to neighbors.
- Full/Incremental Update:
  - Full Update: Send all routing information from own table.
  - Incremental Update: Send only entries that has changed. (Make it fit into one single packet)

DSDV (New Node)

2. Insert entry for D with sequence number D-000
Then immediately broadcast own table

1. D broadcast for first time
Send Sequence number D-000



Dest.	Next	Metric	Seq.
A	A	0	A-550
B	B	1	B-104
C	B	2	C-590

Dest.	Next	Metric	Seq.
A	A	1	A-550
B	B	0	B-104
C	C	1	C-590

Dest.	Next	Metric	Seq.
A	B	2	A-550
B	B	1	B-104
C	C	0	C-590
D	D	1	D-000

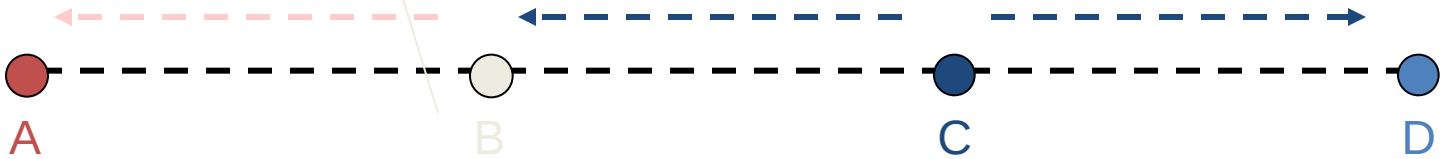
DSDV (New Node cont.)

3. C increases its sequence number to C-592 then broadcasts its new table.

4. B gets this new information and updates its table…….

(A, 2, A-550)
(B, 1, B-102)
(C, 0, C-592)
(D, 1, D-000)

(A, 2, A-550)
(B, 1, B-102)
(C, 0, C-592)
(D, 1, D-000)



Dest.	Next	Metric	Seq.
A	A	0	A-550
B	B	1	B-104
C	B	2	C-590

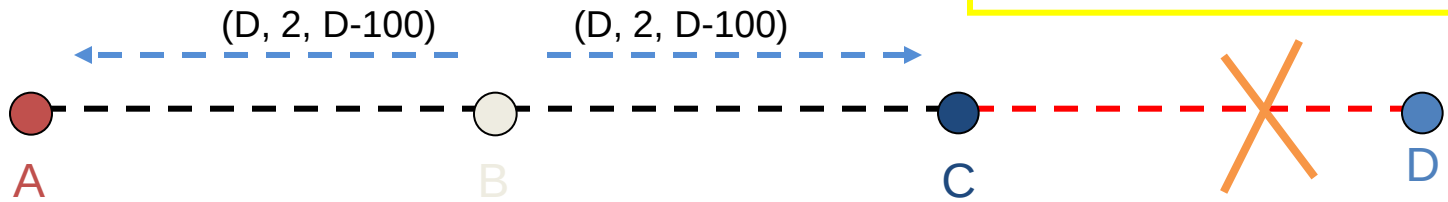
Dest.	Next	Metric	Seq.
A	A	1	A-550
B	B	0	B-102
C	C	1	C-592
D	C	2	D-000

Dest.	Next	Metric	Seq.
A	B	2	A-550
B	B	1	B-102
C	C	0	C-592
D	D	1	D-000

DSDV (No loops, No count to infinity)

2. B does its broadcast
-> no affect on C (C knows that B has stale information because C has higher seq. number for destination D)
-> no loop -> no count to infinity

1. Node C detects broken Link:
-> Increase Seq. Nr. by 1
(only case where not the destination sets the sequence number -> odd number)



Dest.	Next	Metric	Seq.
...	...	...	
D	B	3	D-100

Dest.c	Next	Metric	Seq.
...	...	...	
D	C	2	D-100

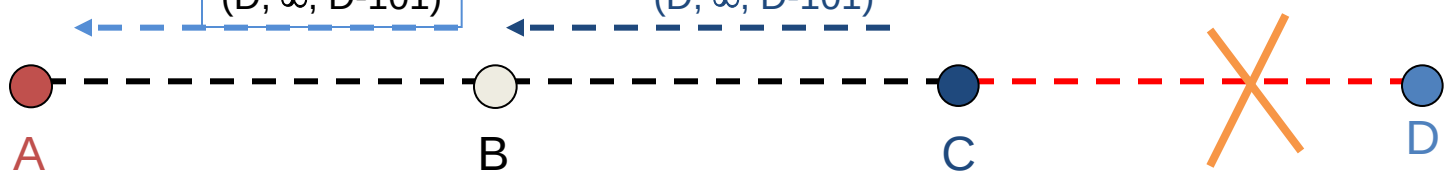
Dest.	Next	Metric	Seq.
...	...	...	
D	D	∞	D-101

DSDV (Immediate Advertisement)

3. Immediate propagation B to A:
(update information has higher Seq. Nr. -> replace table entry)

2. Immediate propagation C to B:
(update information has higher Seq. Nr. -> replace table entry)

1. Node C detects broken Link:
-> Increase Seq. Nr. by 1
(only case where not the destination sets the sequence number -> odd number)

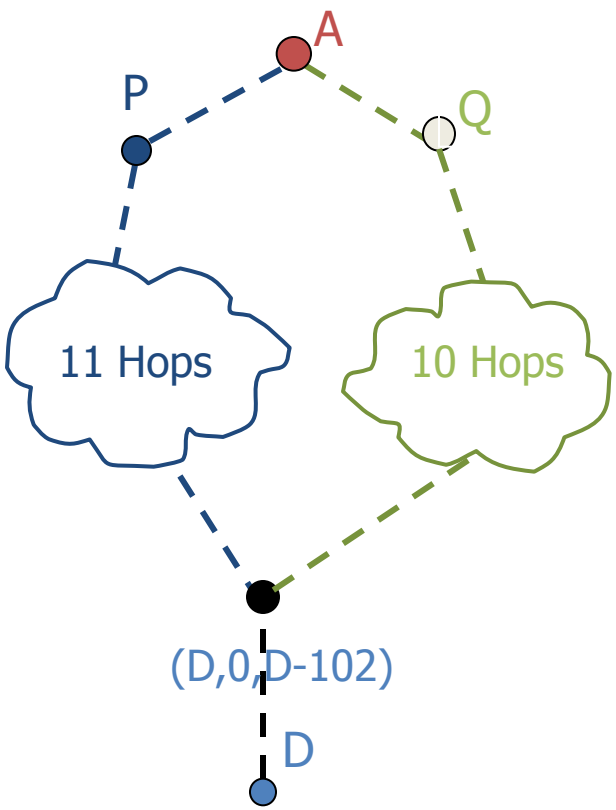


Dest.	Next	Metric	Seq.
...	...	...	...
~~D~~	~~B~~	~~3~~	~~D-100~~
D	B	∞	D-101

Dest.c	Next	Metric	Seq.
...	...	...	...
~~D~~	~~C~~	~~2~~	~~D-100~~
D	C	∞	D-101

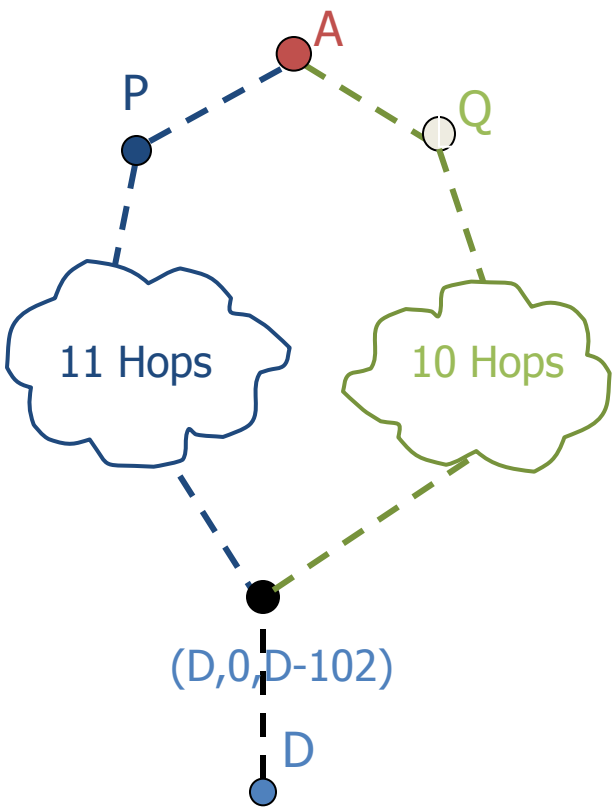
Dest.	Next	Metric	Seq.
...	...	...	
~~D~~	~~D~~	~~1~~	~~D-100~~
D	D	∞	D-101

DSDV (Problem of Fluctuations)



- What are Fluctuations
  - Entry for D in A: [D, Q, 14, D-100]
  - D makes Broadcast with Seq. Nr. D-102
  - A receives from P Update (D, 15, D-102) -> Entry for D in A: [D, P, 15, D-102] A must propagate this route immediately.
  - A receives from Q Update (D, 14, D-102) -> Entry for D in A: [D, Q, 14, D-102]
  - A must propagate this route immediately.
- This can happen every time D or any other node does its broadcast and lead to unnecessary route advertisements in the network, so called fluctuations.

DSDV (Damping Fluctuations)



- How to damp fluctuations
  - Record last and avg. Settling Time of every Route in a separate table. (Stable Data) Settling Time = Time between arrival of first route and the best route with a given seq. nr.
  - A still must update his routing table on the first arrival of a route with a newer seq. nr., but he can wait to advertising it. Time to wait is proposed to be 2*(avg. Settling Time).
  - Like this fluctuations in larger networks can be damped to avoid unnecessary advertisement, thus saving bandwidth.

DSR General

- **Route discovery**
  - Is the mechanism by which a source node S, obtains a route to a destination D
  - Used only when S attempt to send a packet to D and does not already knows a route to D

DSR General

- **Route maintainance**
  - Is the mechanism by which source node S is able to detect if the network topology has changed and can no longer use its route to D
  - If S knows another route to D, use it
  - Else invoke route discovery process again to find a new route
  - Used <u>only</u> when S wants to send a packet to D

DSR General

- Each mechanism operate entirely on demand
- DSR requires no periodic packets of any kind at any level
- Uni-directional and asymmetric routes support (e.g. send a packet to a node D through a route and receive a packet D from another route)

DSR Basic Route Discovery

- When S wants to sent a packet to D
- it places in the header of the packet a source route giving the sequence of hops that the packet should follow on its way to D
- S obtains a suitable source route by searching its route table
- If no route found for D, S initiate the Route Discovery protocol to dynamically find a new route to D

DSR Basic Route Discovery

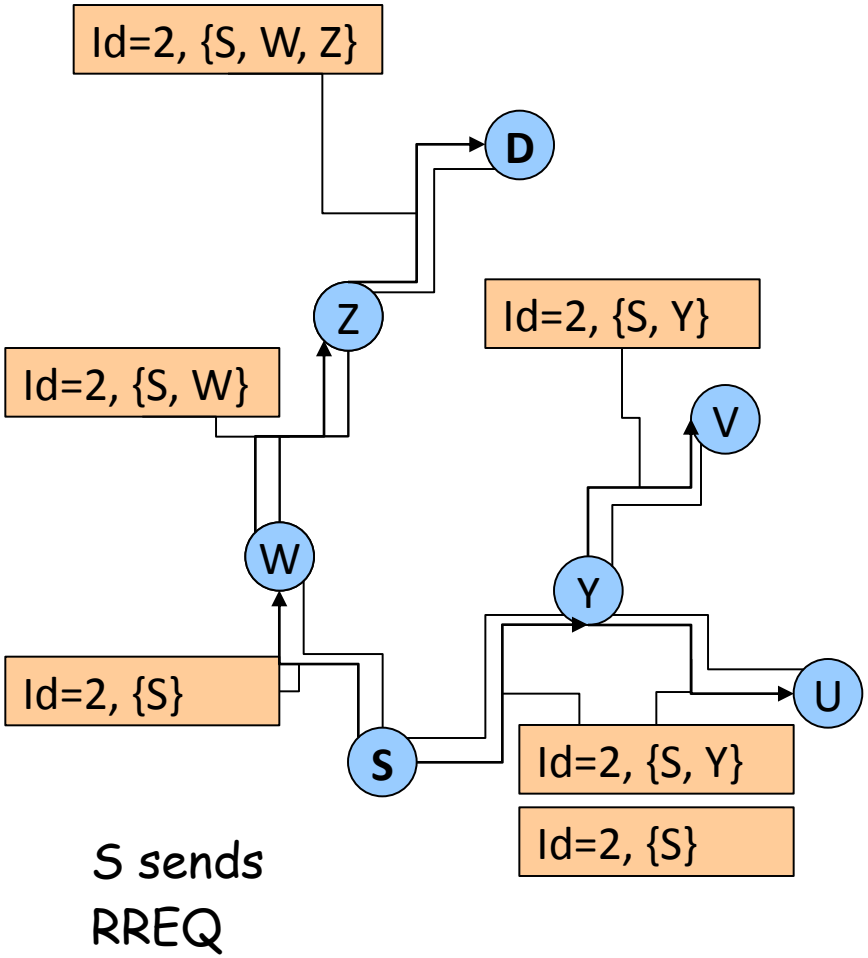
Sender

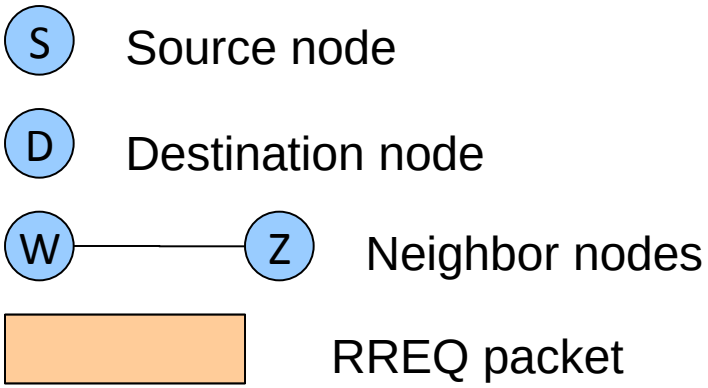
- Broadcasts a Route Request Packet (RREQ)
- RREQ contains a unique **Request ID** and the address of the sender

Receiver

- If this node is the destination node, or has route to the destination send a Route Reply packet (RREP)
- Else if is the source, drop the packet
- Else if is already in the RREQ's route table, drop the packet
- Else append the node address in the RREQ's route table and broadcast the updated RREQ

DSR Basic Route Discovery





DSR Basic Route Discovery

- When a RREQ reaches the destination node, a RREP must be sent back to source
- The destination node
  - Examine its own Route Cache for a route back to source
  - If found, it use this route to send back the RREP
  - Else, the destination node starts a new Route Discovery process to find a route towards source node
- In protocols that require bi-directional links like 802.11, the reversed route list of the RREQ packet can be used, in order to avoid the second Route Discovery

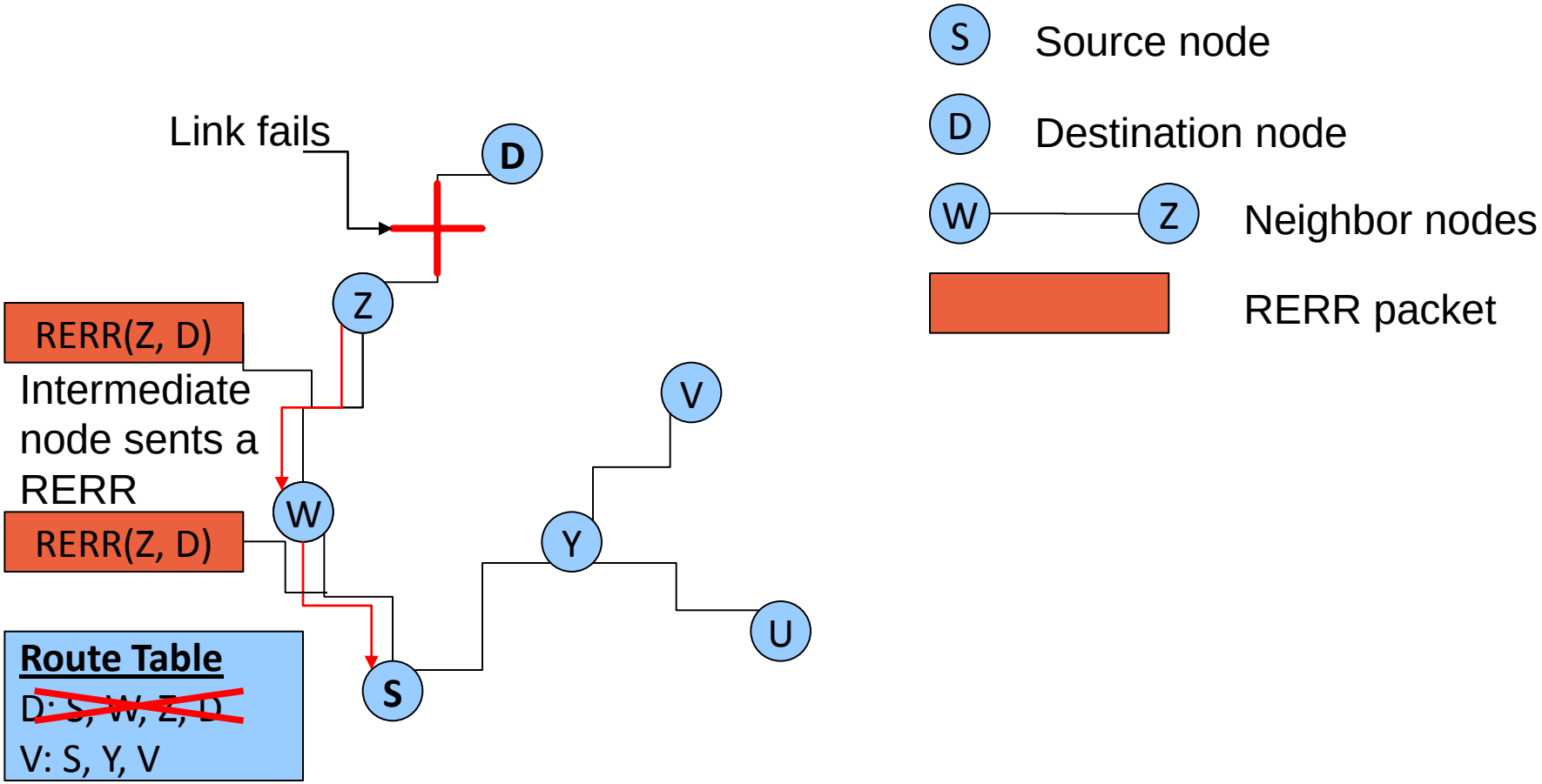
DSR Basic Route Maintenance

- **Each node transmitting a packet**
- is responsible for confirming that the packet has been received by the next hop along the source route
- The confirmation it is done with a standard part of MAC layer (e.g. Link-level ACKs in 802.11)
- If none exists, a DSR-specific software takes the responsibility to sent back an ACK
- When retransmissions of a packet in a node reach a maximum number, a Route Error Packet (RERR) is sent from the node back to the source, identifying the broken link

DSR Basic Route Maintenance

- **The source**
- Removes from the routing table the broken route
- Retransmission of the original packet is a function of upper layers (e.g. TCP)
- It searches the routing table for another route, or start a new Route Discovery process

(DSR) Basic Route Maintenance



AODV Overview

- AODV is a packet routing protocol designed for use in mobile ad hoc networks (MANET)
- Intended for networks that may contain thousands of nodes
- One of a class of demand-driven protocols
  - The route discovery mechanism is invoked only if a route to a destination is not known
- UDP is the transport layer protocol
- Source, destination and next hop are addressed using IP addressing
- Each node maintains a routing table that contains information about reaching destination nodes.
  - Each entry is keyed to a destination node.

AODV Overview

- Routing table size is minimized by only including next hop information, not the entire route to a destination node.
- Sequence numbers for both destination and source are used.
- Managing the sequence number is the key to efficient routing and route maintenance
  - Sequence numbers are used to indicate the relative *freshness* of routing information
  - Updated by an originating node, e.g., at initiation of route discovery or a route reply.
  - Observed by other nodes to determine freshness.

AODV Overview

- The basic message set consists of:
  - RREQ – Route request
  - RREP – Route reply
  - RERR – Route error
  - HELLO – For link status monitoring

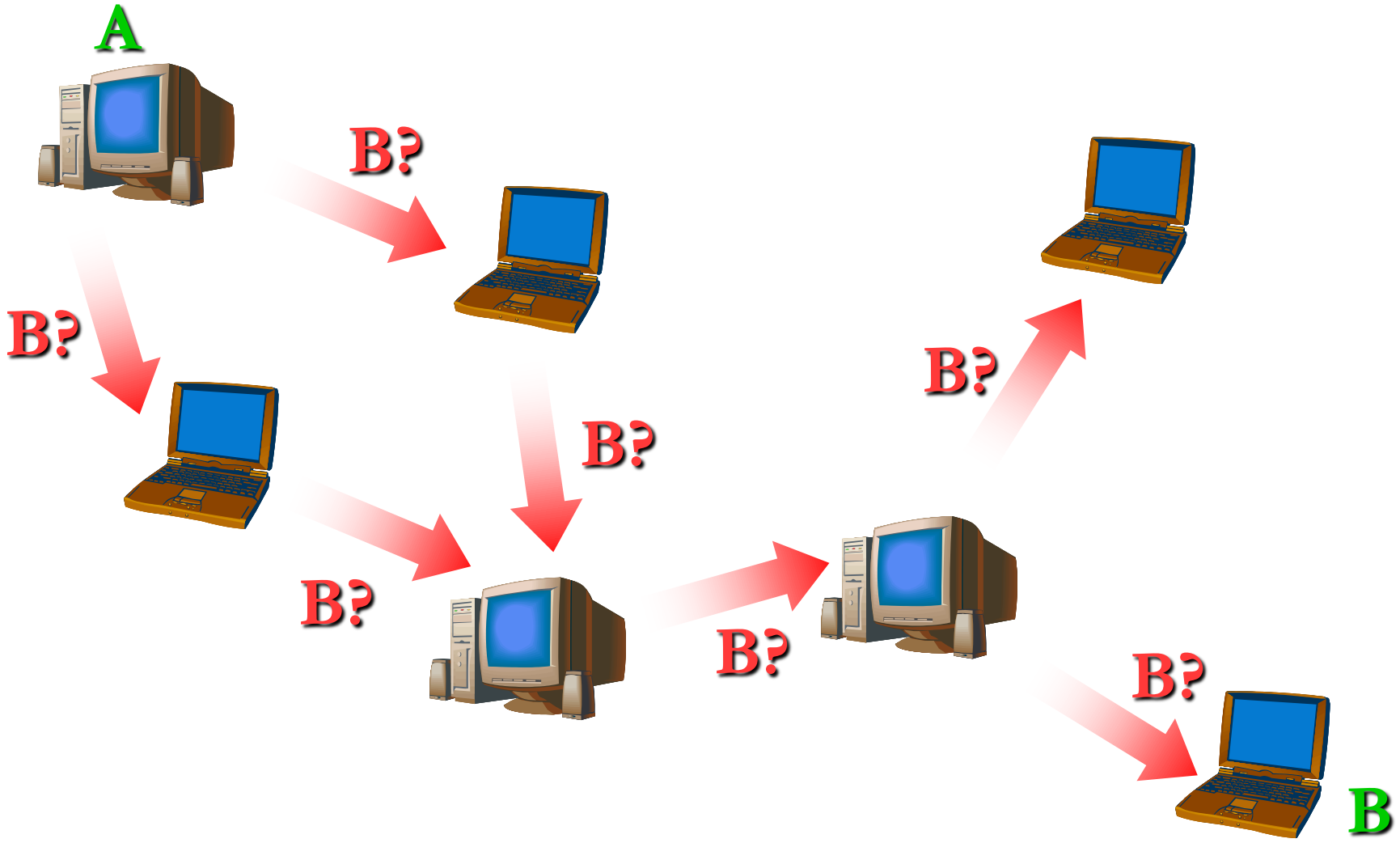
Routing Table Fields

- Destination IP address
- Destination Sequence Number
- Valid Destination Sequence Number Flag
- Other state and routing flags
- Network Interface
- Hop Count (needed to reach destination)
- Next Hop
- Precursor List
- Lifetime (route expiration or deletion time)

AODV Operation – Message Types

- RREQ Messages
  - While communication routes between nodes are valid, AODV does not play any role.
  - A RREQ message is broadcasted when a node needs to discover a route to a destination.
  - As a RREQ propagates through the network, intermediate nodes use it to update their routing tables (in the direction of the source node).
  - The RREQ also contains the most recent sequence number for the destination.
  - A valid destination route must have a sequence number at least as great as that contained in the RREQ.

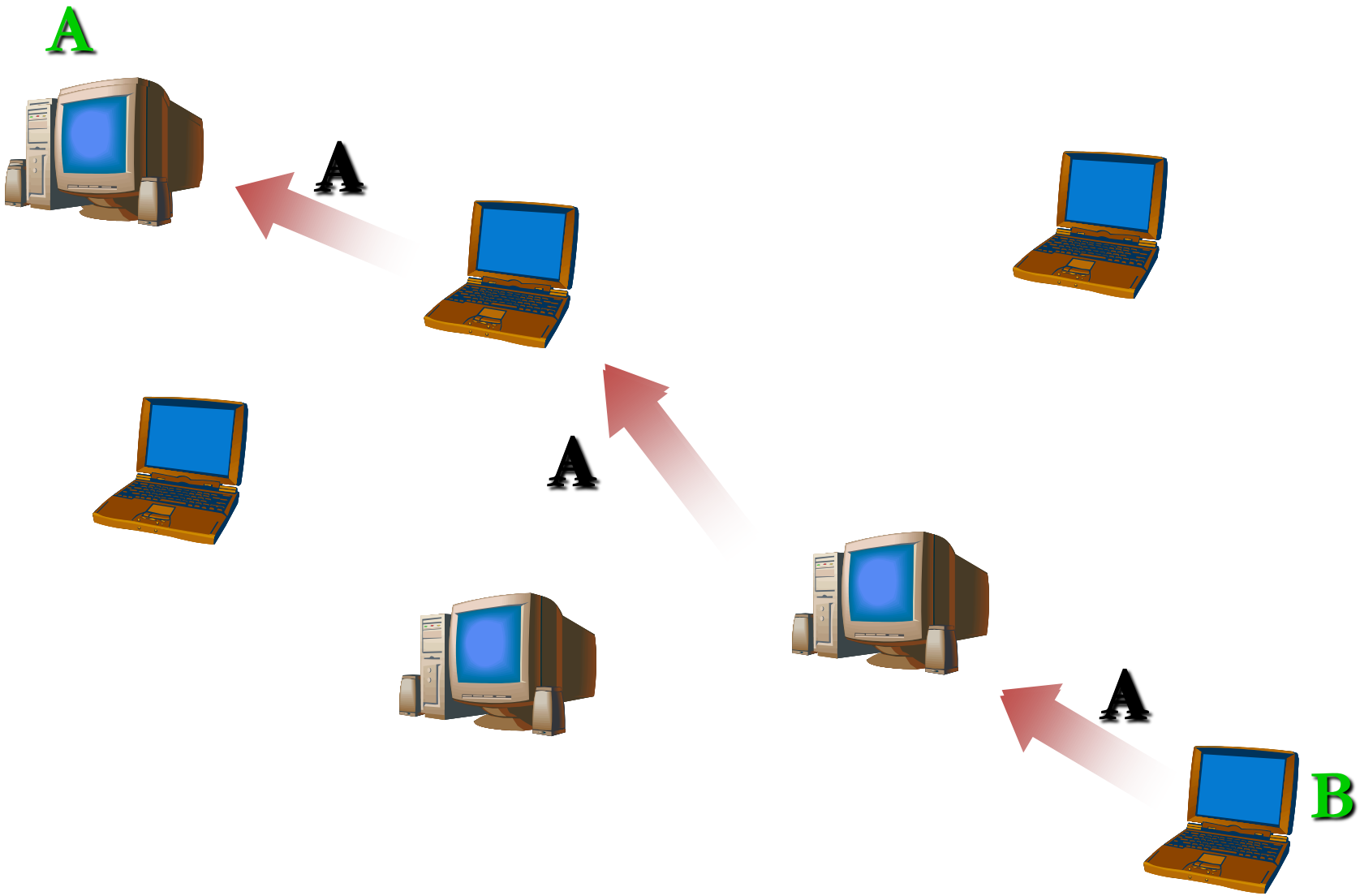
RREQ Message



AODV Operation – Message Types

- RREP Messages
  - When a RREQ reaches a destination node, the destination route is made available by unicasting a RREP back to the source route.
  - A node generates a RREP if:
    - It is itself the destination.
    - It has an active route to the destination. Ex: an intermediate node may also respond with an RREP if it has a "fresh enough" route to the destination.
  - As the RREP propagates back to the source node, intermediate nodes update their routing tables (in the direction of the destination node).

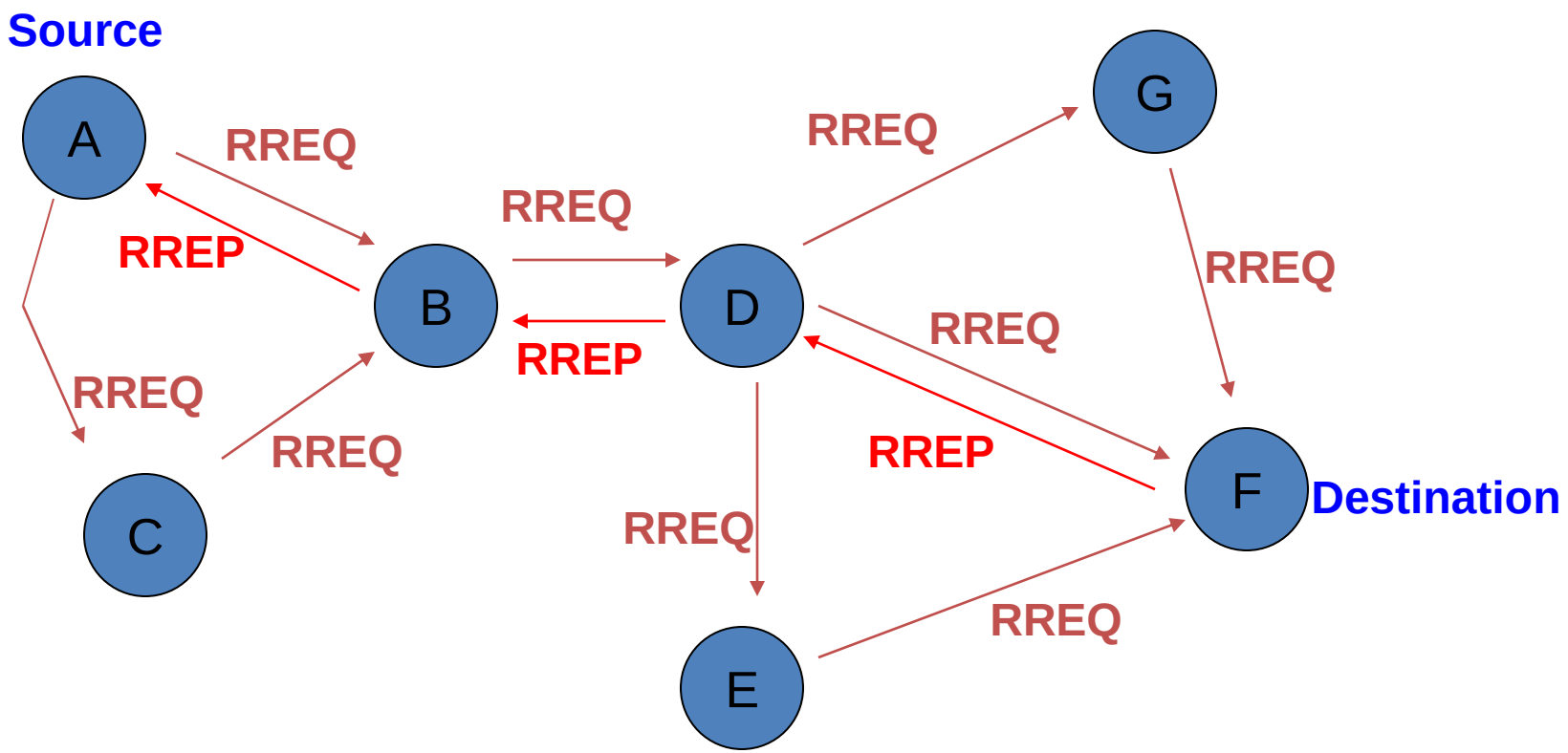
RREP Message



AODV Operation – Message Types

- RERR Messages
  - This message is broadcast for broken links
  - Generated directly by a node or passed on when received from another node
- Hello Messages
  - Hello Message = RREP with TTL = 1
  - This message is used for broadcasting connectivity information.
  - A node should use Hello messages only if it is part of an active route.

Message routing



Congestion Handling

- One method that AODV handle congestion is:
  - If the source node receives no RREP from the destination, it may broadcast another RREQ, up to a maximum of RREQ_RETRIES.
  - For each additional attempt that a source node tried to broadcast RREQ, the waiting time for the RREP is multiplied by 2.
- DSR is not capable of handling congestion.

Congestion Handling

- Other possible methods to improve AODV congestion handling:
  - A route may predict when congestion is about to occur and try to avoid it by reduce the transmission rate.
  - Schedule the requests so that it will not overload the network.

AODV Routing

- There are two phases
  - Route Discovery
  - Route Maintenance
- Each node maintains a routing table with knowledge about the network.
- AODV deals with route table management.
- Route information maintained even for short lived routes – reverse pointers.

Entries in Routing Table

- Destination IP Address
- Destination Sequence Number
- Valid Destination Sequence Number flag
- Other state and routing flags (e.g., valid, invalid, repairable, being repaired)
- Network Interface
- Hop Count (number of hops needed to reach destination)
- Next Hop
- List of Precursors
- Lifetime (expiration or deletion time of the route)
- DSR maintains additional table entries, causing a larger memory overhead

Discovery

- Broadcast RREQ messages.
- Intermediate nodes update their routing table
- Forward the RREQ if it is not the destination.
- Maintain back-pointer to the originator.
- Destination generates RREQ message.
- RREQ sent back to source using the reverse pointer set up by the intermediate nodes.
- RREQ reaches destination, communication starts.

Algorithm for Discovery

- @Originator: If a route to the destination is available, start sending data. Else generate a RREQ packet. Increment the RREQID by 1. Increment the sequence number by 1.Destination IP address, currently available sequence number included.
- @Intermediate Node: Generate route reply, if a 'fresh enough' route is a valid route entry for the destination whose associated sequence number is at least as great as that contained in the RREQ. Change the sequence number of the destination node if stale, increment the hop count by 1 and forward.
- @Destination: 1.Increment sequence number of the destination. 2.Generate a RREQ message and sent back to Originator.

Discovery

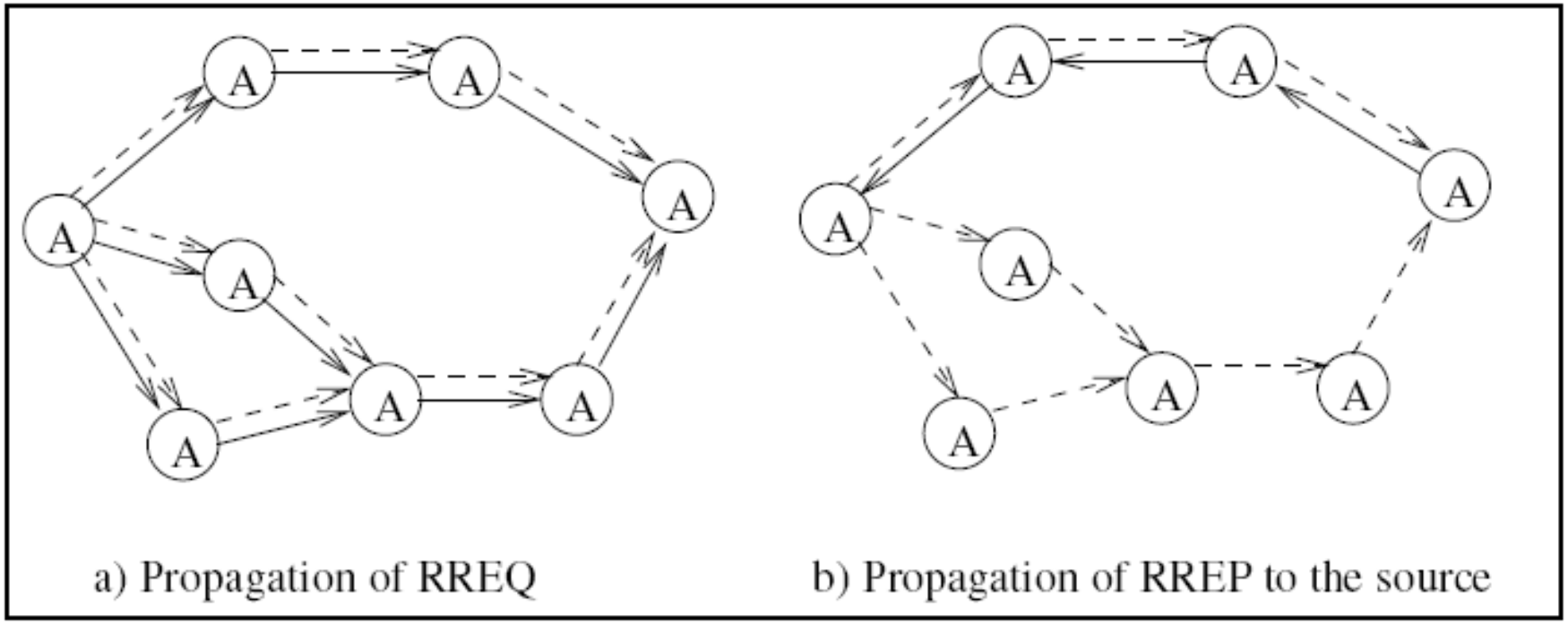


Figure 6: AODV route discovery.

Maintenance

- Hello messages broadcast by *active* nodes periodically HELLO_INTERVAL.
- No hello message from a neighbor in DELETE_PERIOD, link failure identified.
- A local route repair to that next hop initiated.
- After a timeout ,error propagated both to originator and destination.
- Entries based on the node invalidated.

Information "Freshness" Assured

- Each originating node maintains a monotonically increasing sequence number.
- Used by other nodes to determine the freshness of the information.
- Every nodes routing table contains the latest information available about the sequence number for the IP address of the destination node for which the routing information is maintained.
  - Updated whenever a node receives new information about the sequence number from RREQ, RREP, or RERR messages received related to that destination.

Information "Freshness" Assured

- AODV depends on each node in the network to own and maintain its destination sequence number.
- A destination node increments its own sequence number immediately before it originates a route discovery
- A destination node increments its own sequence number immediately before it originates a RREP in response to a RREQ
- The node treats its sequence number as an unsigned number when incrementing accomplishing sequence number rollover.
- Destination information is assured by comparing the sequence number of the incoming AODV message with its sequence number for that destination.

RERR Messages

- Message is broadcasted when
  - A node detects that a link with adjacent neighbor is broken (destination no longer reachable).
  - If it gets a data packet destined to a node for which it does not have an active route and is not repairing.
  - If it receives a RERR from a neighbor for one or more active routes.

RERR Processing

- Build Affected Destination Listing
  I. List unreachable destinations containing unreachable neighbor & destination using unreachable as next hop
  II. Only one unreachable destination, which node already has.
  III. List of nodes where RERR is next hop
- Update information
- Transmit RERR for each item listed

RERR – information update

- Destination Sequence #
  - Update sequence # for case i and ii
  - Copy sequence # for case iii
- Invalidate route entry
- Update Lifetime field as (current time + DELETE_PERIOD)
- Only now may route entry be deleted

RERR message transmission

- Unicast
  - A node detects that a link with adjacent neighbor is broken (destination no longer reachable).
  - Send RERR to single recipient
  - If it gets a data packet destined to a node for which it does not have an active route and is not repairing.
  - If it receives a RERR from a neighbor for one or more active routes.
- Unicast iterative
  - Send RERR to a number of recipients individually
- Broadcast
  - Notify multiple recipients simultaneously
  - Broadcast via 255.255.255.255 TTL = 1

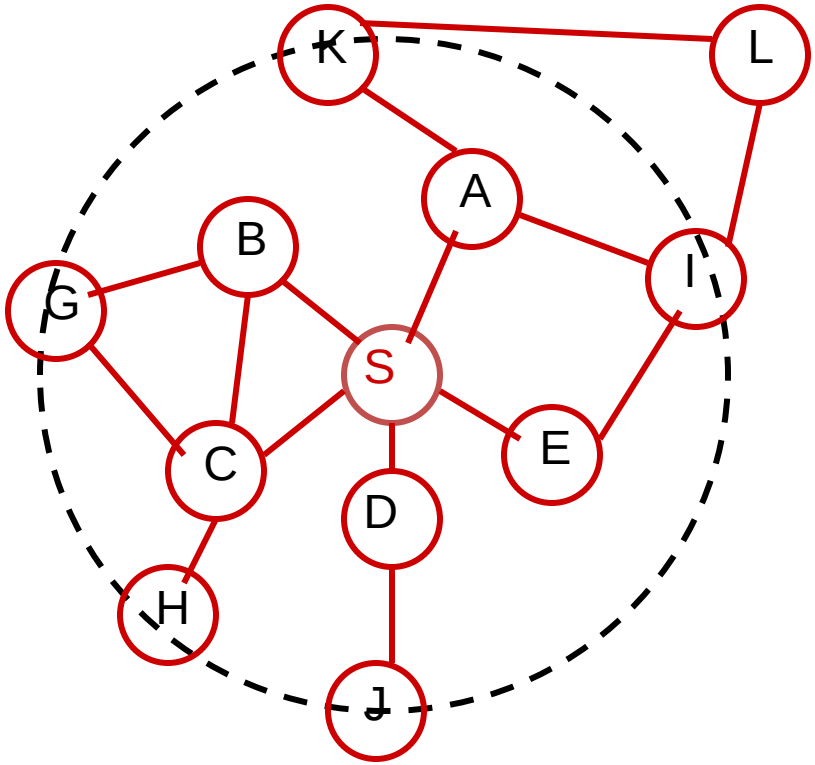
A Combined Protocol

- It is possible to exploit the good features of both reactive and proactive protcols and the **Zone routing protocol** does that.
- The proactive part of the protocol is restricted to a small neighbourhood of a node and the reactive part is used for routing across the network.
- This reduces latency in route discovery and reduces the number of control messages as well.

Routing Zones

- Each node S in the network has a routing zone. This is the proactive zone for S as S collects information about its routing zone in the manner of the DSDV protocol.
- If the radius of the routing zone is k, each node in the zone can be reached within k hops from S.
- The minimum distance of a peripheral node from S is k (the radius).

A Routing Zone



- All nodes except L are in the routing zone of S with radius 2.

Nodes in a Routing Zone

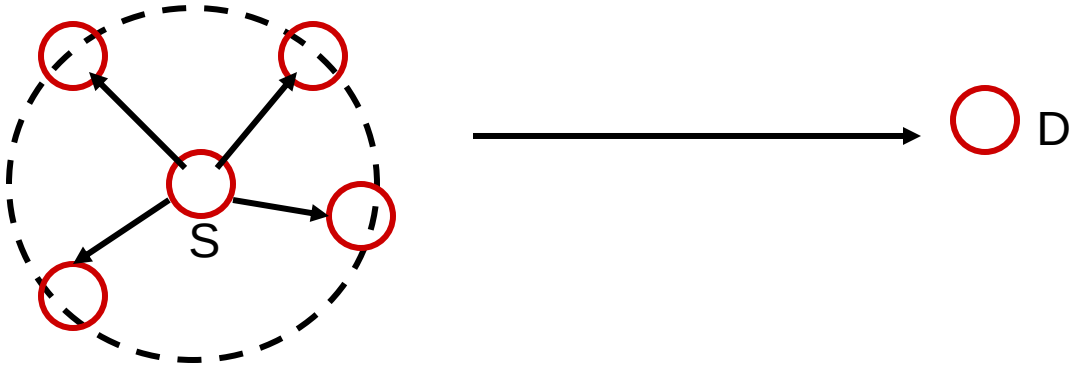
- The coverage of a node´s trasmitter is the set of nodes in direct communication with the node. These are also called neighbours.
- In other words, the neighbours of a node are the nodes which are one hop away.
- For S, if the radius of the routing zone is k, the zone includes all the nodes which are k-hops away.

Neighbour Discovery Protocol

- Like other ad hoc routing protocols, each node executes ZRP to know its current neighbours.
- Each node transmits a hello message at regular intervals to all nodes within its transmission range.
- If a node P does not receive a hello message from a previously known neighbour Q, P removes Q from its list of neighbours.

Basic Strategy in ZRP

- The routing in ZRP is divided into two parts
  - Intrazone routing : First, the packet is sent within the routing zone of the source node to reach the peripheral nodes.
  - Interzone routing : Then the packet is sent from the peripheral nodes towards the destination node.



Intrazone Routing

- Each node collects information about all the nodes in its routing zone proactively. This strategy is similar to a proactive protocol like DSDV.
- Each node maintains a routing table for its routing zone, so that it can find a route to any node in the routing zone from this table.

Intrazone Routing

- In the original ZRP proposal, intrazone routing is done by maintaining a link state table at each node.
- Each node periodically broadcasts a message similar to a hello message kwon as a zone notification message.
- Suppose the zone radius is k for k>1

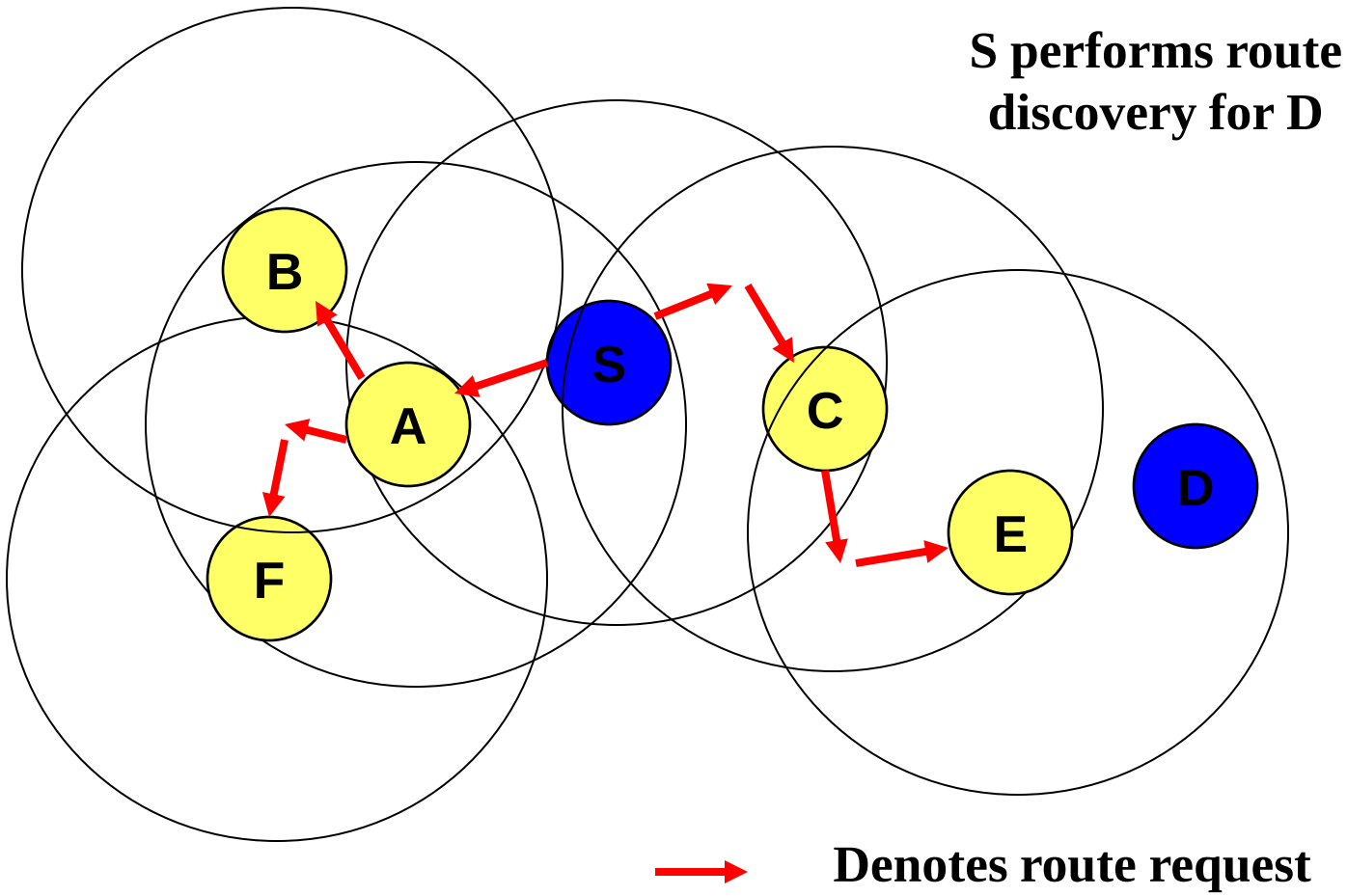
Zone Notification Message

- A hello message dies after one hop, i.e., after reaching a node´s neighbours.
- A zone notification mesage dies after k hops, i.e., after reaching the node´s neighbours at a distance of k hops.
- Each node receiving this message decreases the hop count of the message by 1 and forwards the message to its neighbours.

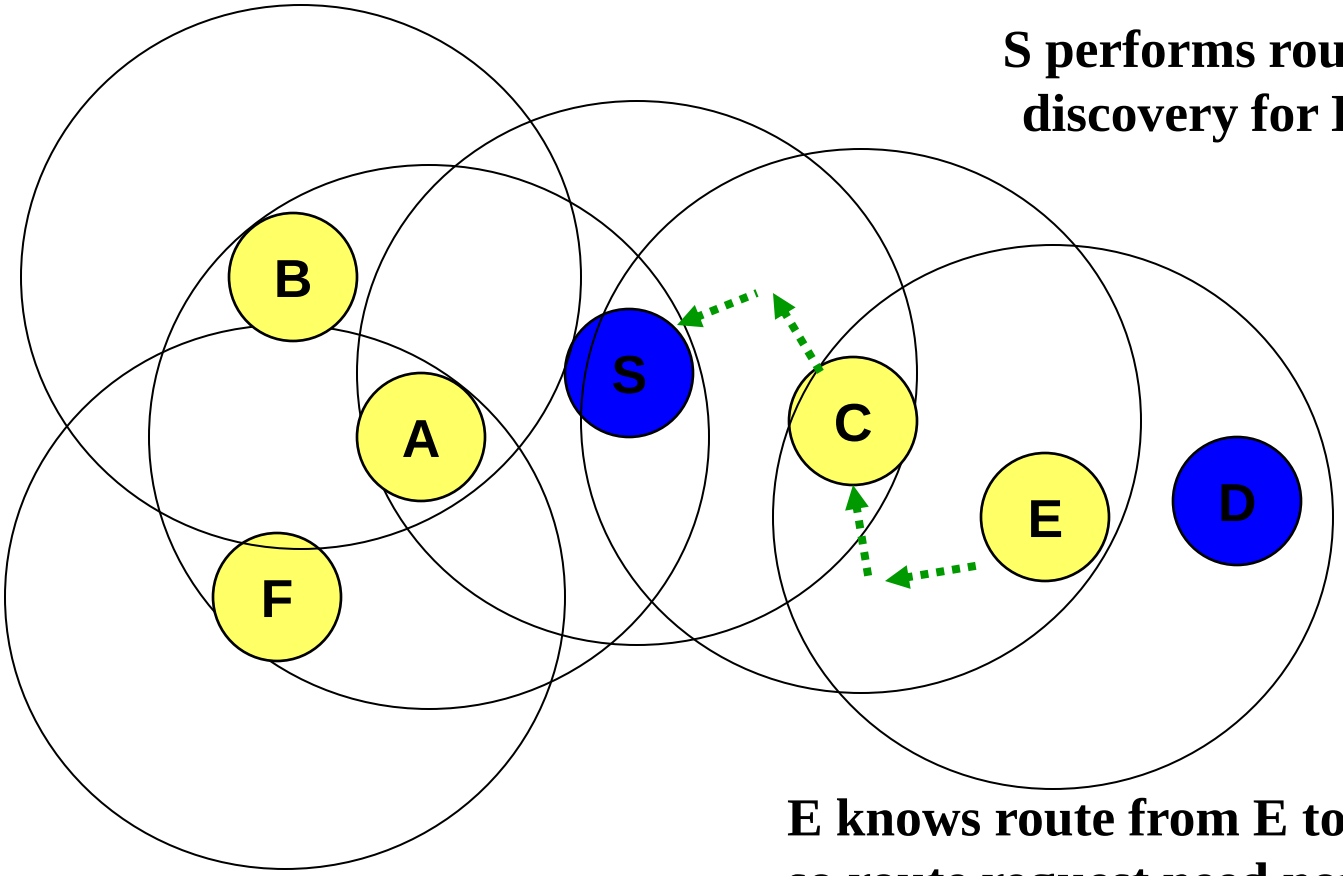
Keeping Track of Nodes in a Routing Zone

- The message is not forwarded any more when the hop count is 0.
- Each node P keeps track of its neighbour Q from whom it received the message through an entry in its link state table.
- P can keep track of all the nodes in its routing zone through its link state table.

ZRP: Example with Zone Radius $K = 2$



ZRP: Example with Zone Radius $K = 2$

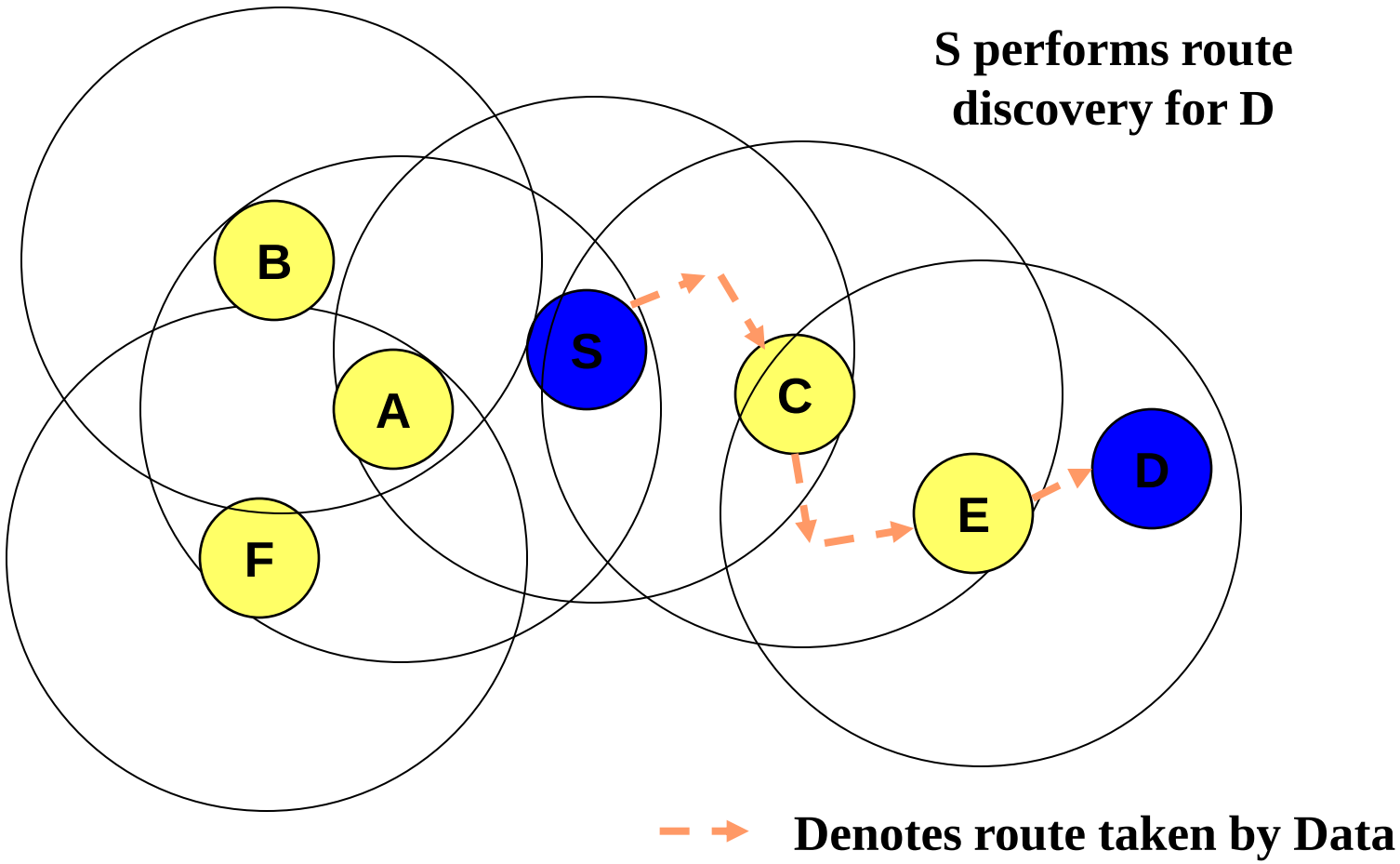


S performs route discovery for D

E knows route from E to D, so route request need not be forwarded to D from E

Denotes route reply

ZRP: Example with Zone Radius $K = 2$



Interzone Routing

- The interzone routing discovers routes to the destination reactively.
- Consider a source (S) and a destination (D). If D is within the routing zone of S, the routing is completed in the intrazone routing phase.
- Otherwise, S sends the packet to the peripheral nodes of its zone through bordercasting.

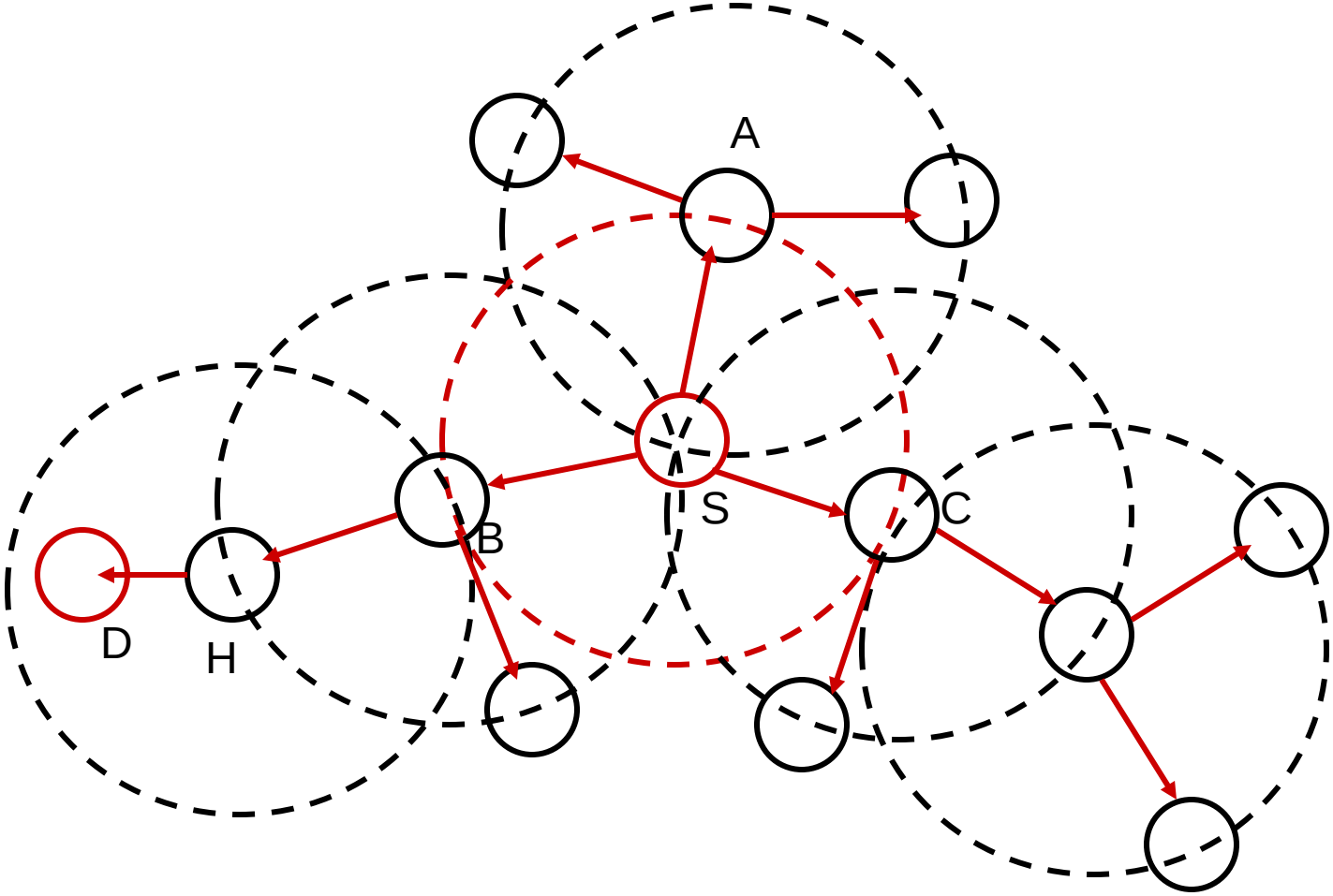
Bordercasting

- The bordercasting to peripheral nodes can be done mainly in two ways
  - By maintaining a multicast tree for the peripheral nodes. S is the root of this tree.
  - Otherwise, S maintains complete routing table for its zone and routes the packet to the peripheral nodes by consulting this routing table.

Interzone Route Discovery

- S sends a route request (RREQ) message to the peripheral nodes of its zone through bordercasting.
- Each peripheral node P executes the same algorithm.
  - First, P checks whether the destination D is within its routing zone and if so, sends the packet to D.
  - Otherwise, P sends the packet to the peripheral nodes of its routing zone through bordercasting.

An Example of Interzone Routing



Route Reply in Interzone Routing

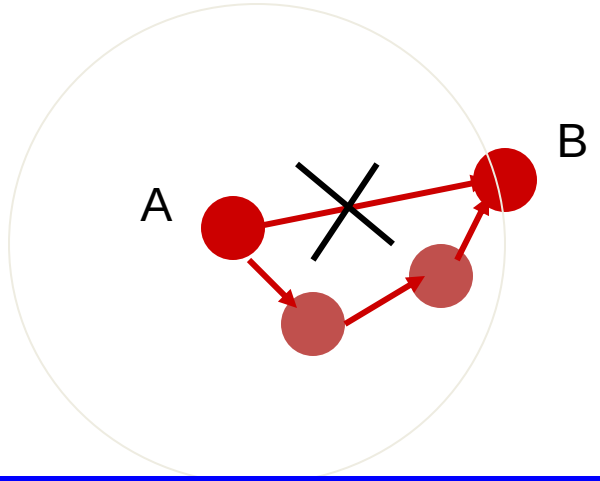
- If a node P finds that the destination D is within its routing zone, P can initiate a route reply.
- Each node appends its address to the RREQ message during the route request phase. This is similar to route request phase in DSR.
- This accumulated address can be used to send the route reply (RREP) back to the source node S.

Route Reply in Interzone Routing

- An alternative strategy is to keep forward and backward links at every node´s routing table similar to the AODV protocol. This helps in keeping the packet size constant.
- A RREQ usually results in more than one RREP and ZRP keeps track of more than one path between S and D. An alternative path is chosen in case one path is broken.

Route Maintenance

- When there is a broken link along an active path between S and D, a local path repair procedure is initiated.
- A broken link is always within the routing zone of some node.



Route Maintenance

- Hence, repairing a broken link requires establishing a new path between two nodes within a routing zone.
- The repair is done by the starting node of the link (node A in the previous diagram) by sending a route repair message to node B within its routing zone.
- This is like a RREQ message from A with B as the destination.

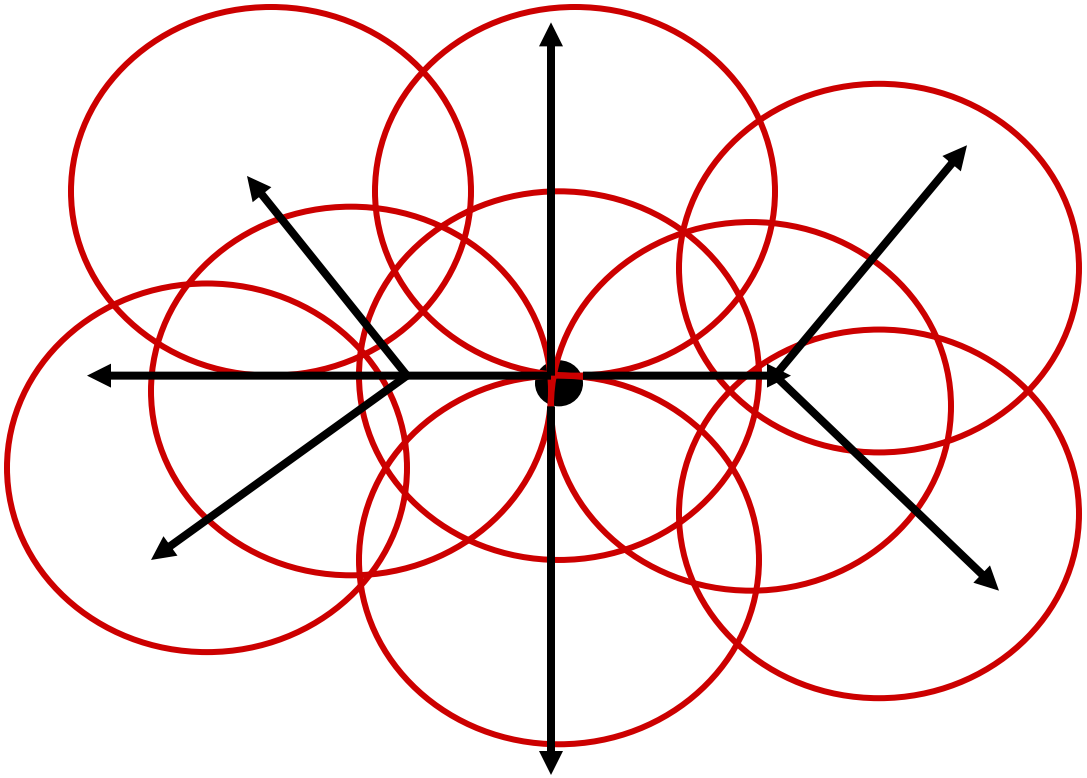
How to Prevent Flooding of the Network

- Interzone routing may generate many copies of the same RREQ message if not directed correctly.
- The RREQ should be steered towards the destination or towards previously unexplored regions of the network.
- Otherwise, the same RREQ message may reach the same nodes many times, causing the flooding of the network.

Routing Zones Overlap Heavily

- Since each node has its own routing zone, the routing zones of neighbouring nodes overlap heavily.
- Since each peripheral node of a zone forwards the RREQ message, the message can reach the same node multiple times without proper control.
- Each node may forward the same RREQ multiple times.

Guiding the Search in InterZone Routing



- The search explores new regions of the network.

Query Forwarding and Termination Strategy

- When a node P receives a RREQ message, P records the message in its list of RREQ messages that it has received.
- If P receives the same RREQ more than once, it does not forward the RREQ the second time onwards.
- Also P can keep track of passing RREQ messages in several different ways.

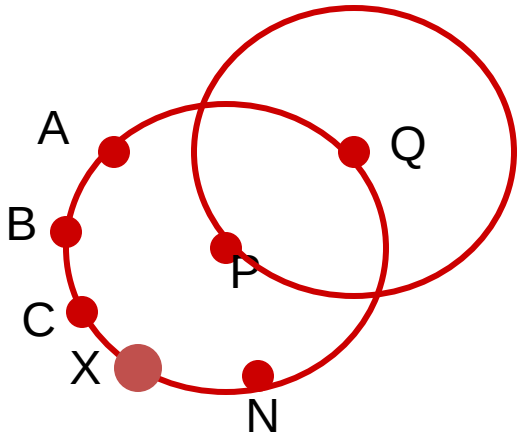
Termination Strategies

- In the promiscuous mode of operation according to IEEE 802.11 standards, a node can overhear passing traffic.
- Also, a node may act as a routing node during bordercasting in the intrazone routing phase.
- Whenever P receives a RREQ message through any of these means, it remembers which routing zone the message is meant for.

Termination Strategies

- Suppose P has a list of nodes A, B,C,...,N such that the RREQ message has already arrived in the routing zones of the nodes A, B, C, ...,N.
- Now P receives a request to forward a RREQ message from another node Q.
- This may happen when P is a peripheral node for the routing zone of Q.

Early Termination of Unnecessary RREQs



- P receives a RREQ from Q since P is a peripheral node for the routing zone of Q.
- P does not bordercast the RREQ to A,B,...,N but only to X which is not in its list.

Evaluation of ZRP

- When the radius of the routing zone is 1, the behaviour of ZRP is like a pure reactive protocol, for example, like DSR.
- When the radius of the routing zone is infinity (or the diameter of the network), ZRP behaves like a pure proactive protocol, for example, like DSDV.
- The optimal zone radius depends on node mobility and route query rates.

Control Traffic

- Control traffic generated by a protocol is the number of overhead packets generated due to route discovery requests.
- In ZRP, control traffic is generated due to interzone and intrazone routing.
- Hello messages transmitted for neighbour discovery are not considered as control traffic since mobility has no effect on it.

Control Traffic for Intrazone Routing

- In the intrazone routing, each node needs to construct the bordercast tree for its zone.
- With a zone radius of r, this requires complete exchange of information over a distance of 2r-1 hops.
- For unbounded networks with a uniform distribution of nodes, this results in O(r^2) intrazone control traffic.

Control Traffic for Intrazone Routing

- However, for a bounded network, the dependence is lower than r^2
- There is no intrazone control traffic when r=1.
- The intrazone control traffic grows fast in practice with increase in zone radius. So, it is important to keep the zone radius small.

Control Traffic for Interzone Routing

- When the zone radius is 1, the control traffic is maximum since ZRP degenerates into flood search.
- In other words, every RREQ message potentially floods the entire network. This is due to the fact that all the neighbours of a node n are its peripheral nodes.
- However, control traffic drops considerably even if the zone radius is just 2.

Control Traffic for Interzone Routing

- The control traffic can be reduced drastically with early query termination, when a RREQ message is prevented from going to the same region of the network multiple times.
- However, the amount of control traffic depends both on node mobility and query rate.
- The performance of ZRP is measured by compairing control traffic with call-to-mobility (CMR) ratio.

Control Traffic for Interzone Routing

- The call-to-mobility ratio (CMR) is the ratio of route query rate to node speed.
- As CMR increases, the number of control messages is reduced by increasing the radius of the routing zones.
- This is because, it is easier to maintain larger routing zones if mobility is low. Hence, route discovery traffic also reduces.

Control Traffic for Interzone Routing

- On the other hand, CMR is low if mobility is high.
- In such a case, the routing zone maintenance becomes very costly and smaller routing zones are better for reducing control traffic.
- An optimally configured ZRP for a CMR of 500 [query/km] produces 70% less traffic than flood searching.

Route Query Response Time

- For a fixed CMR, the route query response time decreases initially with increased zone radius.
- However, after a certain radius, the response time increases with zone radius.
- This is due to the fact that the network takes longer time to settle even with small changes in large routing zones.

Expected advantages from multicast routing

- Providing efficient bandwidth
- Reducing communication cost
- Efficient delivery of data
- Supporting dynamic topology

Technical constraints for multicast routing

- Minimizing network load
- Providing basic support for reliable transmission
- Designing optimal routes
- Providing robustness, efficiency, and adaptability

Classification

- Globally, there are two main categories of multicast routing protocols
- Tree-based protocols
- Mesh-based protocols

Examples of tree-based protocols

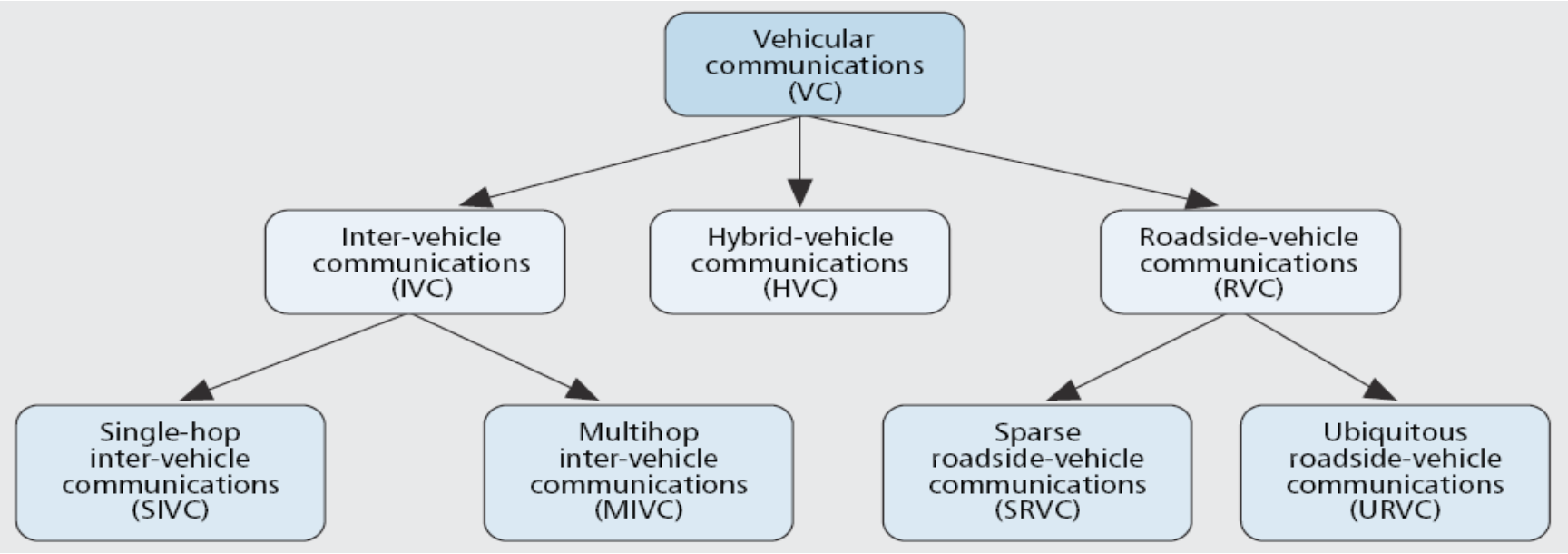
- Multicast Ad hoc On-Demand Distance Vector (MAODV) routing protocol
  - Extends AODV to offer multicast capabilities
  - Builds shared multicast trees on-demand to connect group members
  - Capable of unicast, broadcast, and multicast
- Associativity based Multicast (ABAM) routing protocol
  - Constructed in an attempt to reduce communication overhead and end-to-end delay

An example of mesh-based protocols

- On-Demand Multicast Routing Protocol (ODMRP)
- ODMRP is based on a mesh structure for connecting multicast members using the concept of forwarding group nodes.
- When a data packet reaches a multicast receiver, the receiver creates a Join-Table and broadcasts it to the neighbors.
- Each group member propagates the Join-Table until it reaches the multicast source via the shortest path.
- This process constructs and updates the routes from the source to the receiver, creating a mesh of nodes.

VANETs

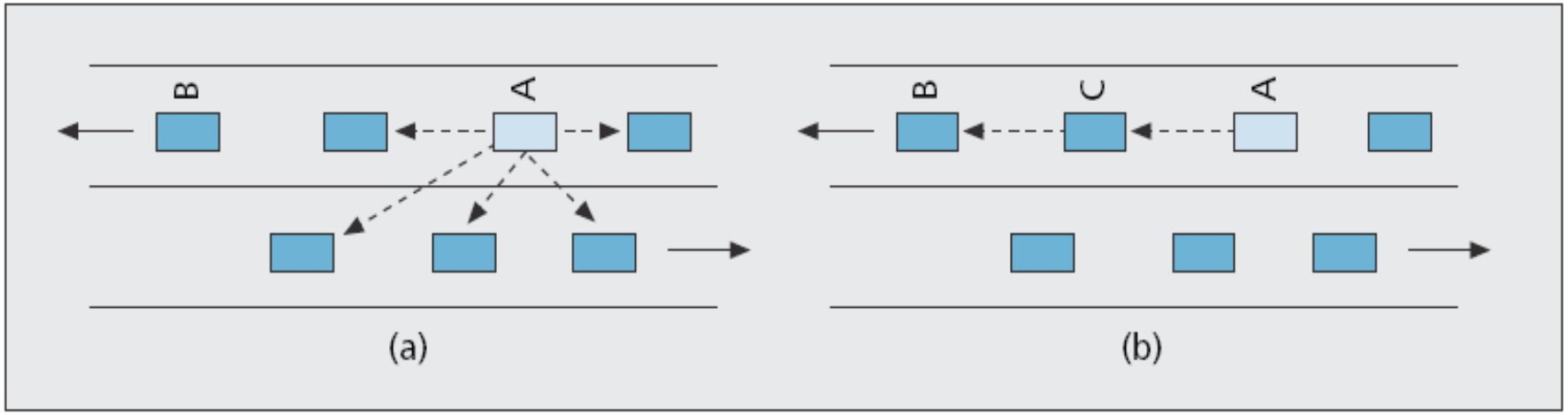
- A VANET (Vehicular Ad hoc NETwork) is a special kind of MANET in which packets are exchanged between mobile nodes (**vehicles**) traveling on constrained paths



Inter-vehicle communication (IVC) Systems

- IVC systems are completely infrastructure-free; only onboard units (OBUs) sometimes also called in-vehicle equipment (**IVE**) are needed.
- Single-hop and multi-hop IVCs (**SIVCs and MIVCs).**
- SIVC systems are useful for applications requiring short-range communications (e.g., lane merging, automatic cruise control)
- MIVC systems are more complex than SIVCs but can also support applications that require long-range communications (e.g., traffic monitoring)

IVC systems



a) Single-hop IVC system

b) multi-hop IVC system

Roadside-to-Vehicle Communication (RVC) Systems

- RVC systems assume that all communications take place between roadside infrastructure (including roadside units [RSUs]) and OBUs.
- Depending on the application, two different types of infrastructure can be distinguished
  - Sparse RVC (**SRVC**) system
  - Ubiquitous RVC (**URVC**) system

RVC Systems –SRVC

- SRVC systems are capable of providing communication services at hot spots.
- A busy intersection scheduling its traffic light, a gas station advertising its existence (and prices), and parking availability at an airport, are examples of applications requiring an SRVC system.
- An SRVC system can be deployed gradually, thus not requiring substantial investments before any available benefits.

RVC Systems -URVC

- A URVC system : providing all roads with high-speed communication would enable applications unavailable with any of the other systems.
- Unfortunately, a URVC system may require considerable investments for providing full (even significant) coverage of existing roadways (especially in large countries like the United States)

Hybrid Vehicular Communication (HVC) Systems

- HVC systems are proposed for extending the range of RVC systems.
- In HVC systems vehicles communicate with roadside infrastructure even when they are not in direct wireless range by using other vehicles as mobile routers.
- An HVC system enables the same applications as an RVC system with a larger transmission range.
- The main advantage is that it requires less roadside infrastructure. However, one disadvantage is that network connectivity may not be guaranteed in scenarios with low vehicle density.

IVC vs. MANET

- MANETs are wireless multihop networks that lack infrastructure, and are decentralized and self-organizing
- IVC systems satisfy all these requirements, and are therefore a special class of MANETs
- There are several characteristics that differentiate IVCs from the common assumptions made in the MANET literature:
  - Applications
  - Addressing
  - Rate of Link Changes
  - Mobility Model
  - Energy Efficiency

IVC vs. MANET

- **Applications**
  - While most MANET articles do not address specific applications, the common assumption in MANET literature is that MANET applications are identical (or similar) to those enabled by the Internet.
  - In contrast, as we show later, IVCs have completely different applications. An important consequence of the difference in the applications is the difference in the addressing modes.

IVC vs. MANET

- **Addressing**
  - Faithful to the Internet model, MANET applications require point-to-point (**unicast) with fixed addressing;** that is, the recipient of a message is another node in the network specified by its IP address.
  - IVC applications often require dissemination of the messages to many nodes (**multicast) that satisfy** some geographical constraints and possibly other criteria (e.g., direction of movement). The need for this addressing mode requires a significantly different routing paradigm.

IVC vs. MANET

- **Rate of Link Changes**
  - In MANETs, the nodes are assumed to have moderate mobility. This assumption allows MANET routing protocols (e.g., Ad Hoc On Demand Distance Vector, AODV) to establish end-to-end paths that are valid for a reasonable amount of time and only occasionally need repairs.
  - In IVC applications, it is shown that due to the **high degree of mobility of the nodes involved, even** multi-hop paths that only use nodes moving in the same direction on a highway have a lifetime comparable to the time needed to discover the path.

IVC vs. MANET

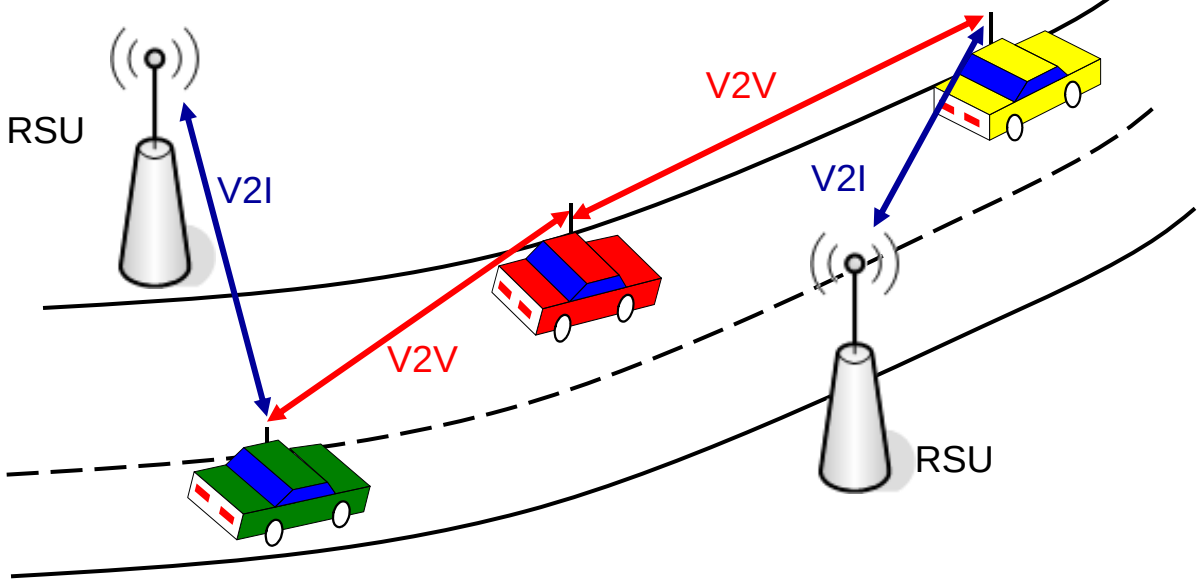
- **Mobility Model**
  - In MANETs, the random waypoint (RWP) is (by far) the most commonly employed mobility model. However, for IVC systems, most existing literature recognized that RWP would be a very poor approximation of real vehicular mobility; instead, detailed vehicular traffic simulators are used.
- **Energy Efficiency**
  - While in MANETs a significant body of literature is concerned with power-efficient protocols, IVC enjoys a practically unlimited power supply.

VANETs

- Like MANETs:
  - They self-organize over an evolving topology
  - They may rely on multi-hop communications
  - They can work without the support of a fixed infrastructure
- Unlike MANETs:
  - They have been conceived for a different set of applications
  - They move at higher speeds (0-40 m/s)
  - They do not have battery and storage constraints

VANETs

- Communication modes:
  - Vehicle-to-Vehicle (V2V) among vehicles
  - Vehicle-to-Infrastructure (V2I), between vehicles and Road-Side Units (RSUs)
  - Vehicle-to-X (V2X), mixed V2V-V2I approach



VANETs Applications

- **Active Road-Safety Applications**
  - To avoid the risk of car accidents: *e.g.*, cooperative collision warning, pre-crash sensing, lane change, traffic violation warning
- **Traffic efficiency and management applications**
  - To optimize flows of vehicles: *e.g.*, enhanced route guidance/navigation, traffic light optimal scheduling, lane merging assistance
- **Comfort and Infotainment applications**
  - To provide the driver with information support and entertainment: *e.g.*, point of interest notification, media downloading, map download and update, parking access, media streaming, voice over IP, multiplayer gaming, web browsing, social networking

VANETs

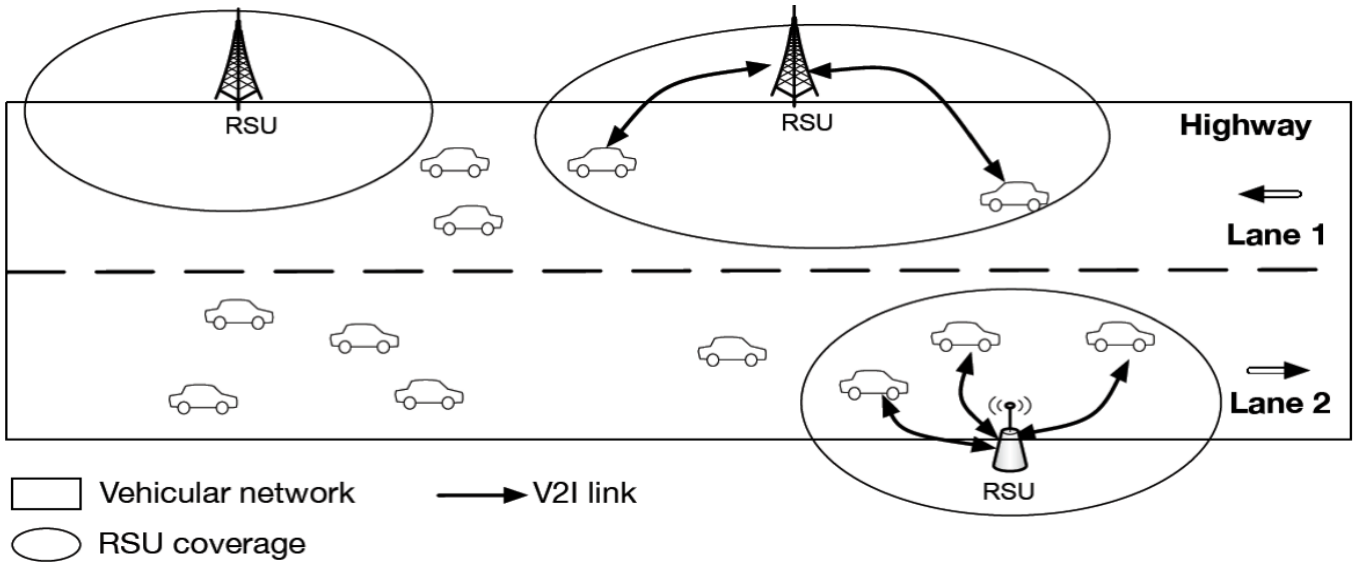
- VANETs applications exhibit very heterogeneous requirements
  - Safety applications require **reliable, low-latency**, and efficient message dissemination
  - Non-safety applications have very **different communication requirements**, from no special real-time requirements of traveler information support applications, to guaranteed Quality-of-Service needs of multimedia and interactive entertainment applications

Connectivity in VANETs

- There are three primary models for interconnecting vehicles based on
  - Network infrastructure
  - Inter-vehicle communications
  - Hybrid configuration

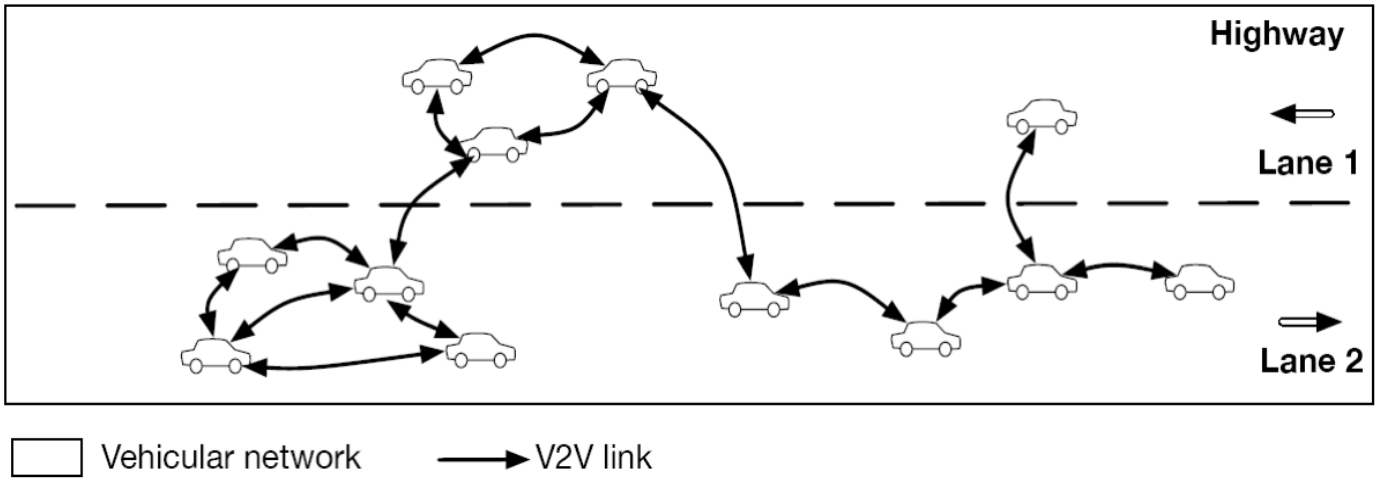
Connectivity in VANETs

- Network infrastructure
  - Vehicles connect to a centralized server or a backbone network such as the Internet, through the road-side infrastructure, *e.g.*, cellular base stations, IEEE 802.11 Access Points, IEEE 802.11*p* RSUs



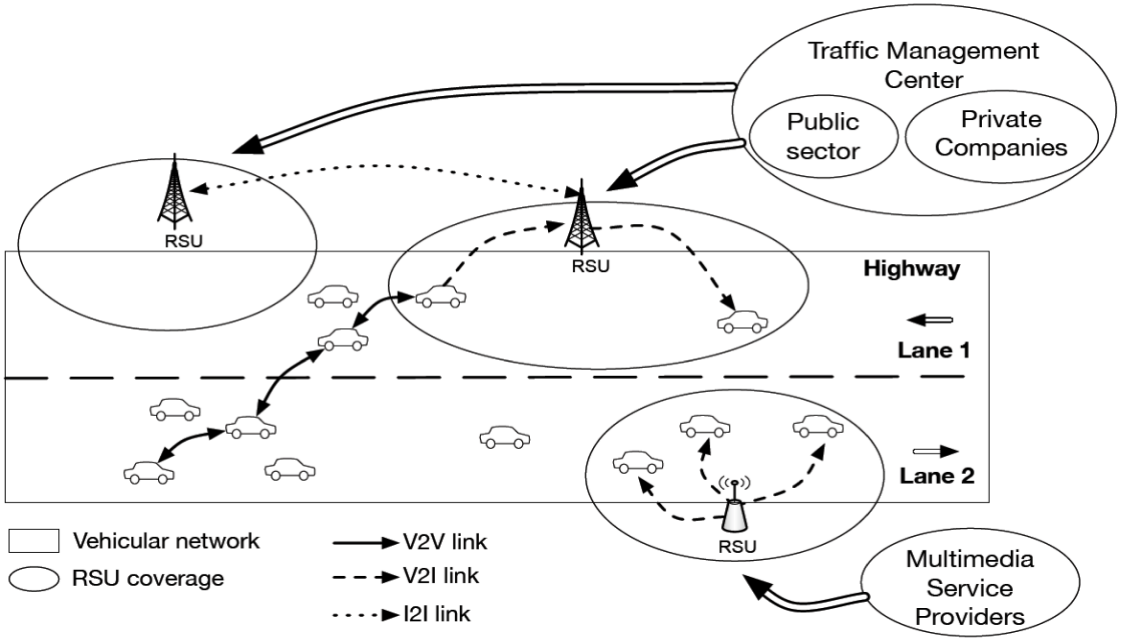
Connectivity in VANETs

- Inter-vehicle communications
  - Use of direct ad-hoc connectivity among vehicles via multihop for applications requiring long-range communications (*e.g.*, traffic monitoring), as well as short-range communications (*e.g.*, lane merging)



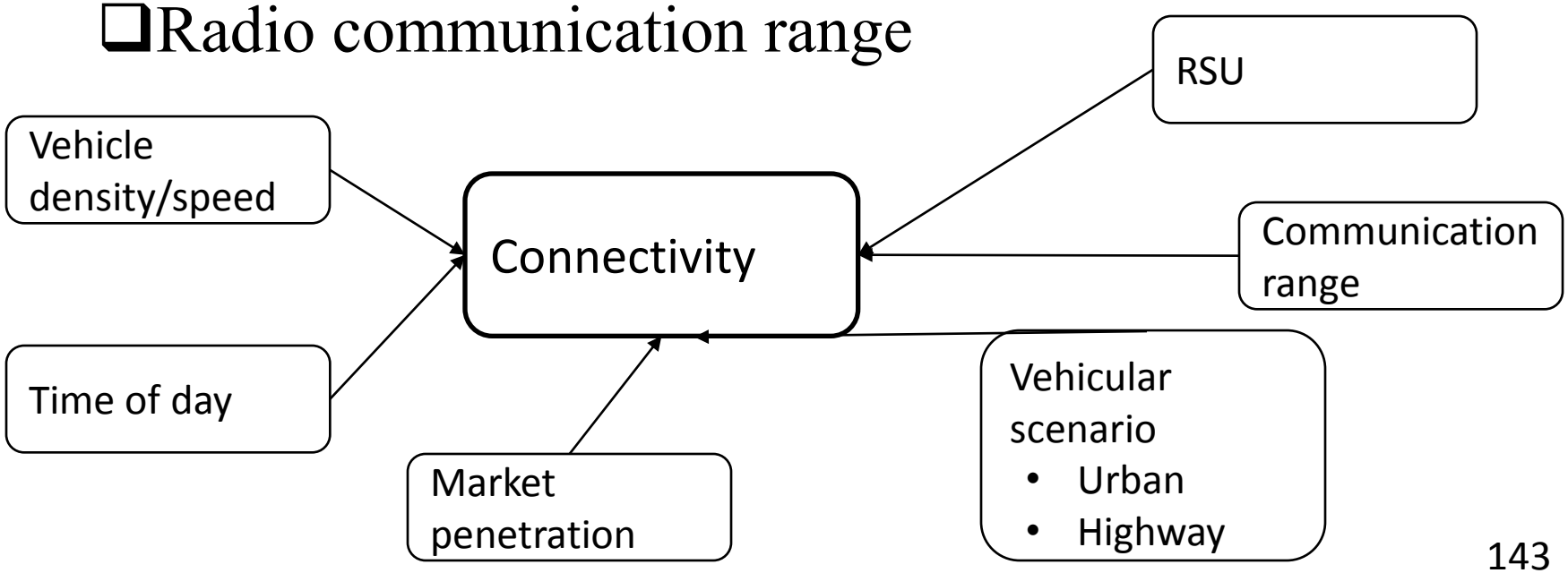
Connectivity in VANETs

- Hybrid configuration
  - Use of a combination of V2V and V2I. Vehicles in range directly connect to the road-side infrastructure, while exploit multi-hop connectivity otherwise



Connectivity in VANETs

- Vehicles' connectivity is determined by a combination of several factors, like:
  - Space and time dynamics of moving vehicles (*i.e.*, vehicle density and speed)
  - Density of RSUs
  - Radio communication range



Improving Connectivity in VANETs

- **Opportunistic approaches** for connectivity support in VANETs
  - Opportunistic contacts, both among vehicles and from vehicles to available RSUs, can be used to instantiate and sustain both safety and non-safety applications
- Opportunistic forwarding is the main technique adopted in DTN
  - In VANETs, **bridging technique** links the partitioning that exists between clusters traveling in the same direction of the roadway

Improving Connectivity in VANETs

- The use of a vehicular grid together with an opportunistic infrastructure placed on the roads guarantees seamless connectivity in dynamic vehicular scenarios
- Hybrid communication paradigms for vehicular networking are used to limit intermittent connectivity
  - Vehicle-to-X (V2X) works in heterogeneous scenarios, where overlapping wireless networks partially cover the vehicular grid. It relies on the concept of multi-hop communication path

Improving Connectivity in VANETs

- Different connectivity phases
  - **Phase 1 (No connectivity)**
    - A vehicle is traveling alone in the vehicular grid (totally-disconnected traffic scenario). The vehicles are completely disconnected
  - **Phase 2 (Short-range connectivity)**
    - A vehicle is traveling in the vehicular grid and forming a cluster with other vehicles. Only V2V connectivity is available
  - **Phase 3 (Long-range connectivity)**
    - A vehicle is traveling in the vehicular grid with available neighboring RSUs. Only V2I connectivity is assumed to be available

Examples

Applications for VANETs

- Public Safety Applications
- Traffic Management Applications
- Traffic Coordination and Assistance Applications
- Traveller Information Support Applications
- Comfort Applications
- Air pollution emission measurement and reduction
- Law enforcement
- Broadband services

Problems in MANET

- Routing
- Security and Reliability
- Quality of Service
- Internetworking
- Power Consumption

SECURITY

- A major issue in Mobile ad-hoc network is "SECURITY".
- Two approaches in protecting mobile ad-hoc networks
  - **Reactive approach:** Seeks to detect security threats and react accordingly.
  - **Proactive approach:** Attempts to prevent an attacker from launching attacks through various cryptographic techniques.

Issues

- Secure Multicasting
- Secure routing
- Privacy-aware Routing
- Key management
- Intrusion detection System

Issues Contd..

- **Secure multicasting:** Is a communication method where a single data packet can be transmitted from a sender and replicated to a set of receivers.
- **Secure routing:** Most MANET routing protocols are vulnerable to attacks that can freeze the whole network. Need some solutions that work even if some nodes compromised.
- **Privacy-aware Routing:** Building routing protocols that prevent intermediate nodes from performing traffic analysis.
- Schemes for minimizing size of crypto-tags(digital signatures) are needed.

Issues Contd..

- **Key Management**
  - security goals in MANET are mainly achieved through trusted Certificate Authority (CA)
  - compromised CA can easily damage the entire network.
- **Intrusion detection and response schemes**: Anomaly detection is difficult in MANETs (ex: types of attacks and their source). collaborative IDS schemes are needed.

Security Goals

- Authentication
- Confidentiality
- Integrity
- Non-repudiation
- Availability
- Detection and Isolation

Security Goals

- Authentication: A node must know the identity of the peer node it is communicating with. Without authentication, an attacker could gain sensitive information and interfere with other nodes
- Confidentiality: Ensures certain information is never disclosed to unauthorized entities.
- Integrity: Message being transmitted is never corrupted.
- Non-Repudiation: The sender cannot later deny sending the information and the receiver cannot deny the reception.
- Availability: Nodes should be available for communication at all times. A node need continue to provide services despite attacks.
  - E.g.: Key management service.
- Detection and Isolation: Require the protocol can identify misbehaving nodes and render them unable to interfere with routing.

IDS-MANET

- IDS: Intrusion detection System which is used to detect and report the malicious activity in ad hoc networks.
- Ex: Detecting critical nodes using IDS
- Intrusion Detection System (IDS) can collect and analyze audit data for the entire network.
- Critical node is a node whose failure or malicious behavior disconnects or significantly degrades the performance of the network.
- Packets may be dropped due to network congestion or because a malicious node is not faithfully executing a routing algorithm.
- Researchers have proposed a number of collaborative IDS systems.
- Some of the schemes are neighbor-monitoring, trust-building, and cluster-based voting schemes which are used to detect and report the malicious activity in ad hoc networks.

Questions ?